

Honest Machine Learning on a Small Clinical Microarray Cohort: Cross-Validated Signal in GSE2034, a 15-Gene Stability Panel Falsified on Two Independent Cohorts, and the Limits of Bootstrap-Stability Panel Discovery

MEGA-PROGRAM-27, Item 16 - computational biology research lane

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Abstract

GSE2034 (286 lymph-node-negative breast cancer patients, 22,283 Affymetrix probes, bone-relapse labels, 24.1% prevalence) is a classic small-n large-p cohort. This paper documents a complete, locked-gate methods arc, with every negative preserved. First negative: a naive 70/30 split logistic model scores AUC 0.537 - an observed weak result, kept in the record. Pivot: 5-fold cross-validation with per-fold feature selection yields logistic AUC 0.632 +/- 0.055 - a modest within-cohort association. Second negative: a co-expression-graph smoothing arm scores 0.629 versus 0.632 plain - no help, reported not buried. Discovery arm: 40 bootstrap-CV refits yield a 15-gene sign-stable candidate panel. THIRD AND FOURTH NEGATIVES, under a pre-registered external-replication rung locked before any replication analysis: the frozen 15-gene sign-locked score FAILS on METABRIC (1,979 patients, 803 relapses; AUC 0.5117, one-sided 95% bootstrap lower bound 0.4778, sign concordance 9/15 - gate requires both arms, primary test fails) and FAILS on SCAN-B GSE96058 (3,409 patients, 353 OS events, declared OS-endpoint arm; AUC 0.5075, lower bound 0.4790, concordance 6/15). The panel is declared NOT externally validated. FIFTH implemented result: the two-cohort pivot (L2 execution deviates from preregistered L1) (cross-cohort stability selection on GSE2034 + METABRIC, test cohorts not used for this selection, with prior scoring disclosed) yields ZERO panel genes - 268 genes are stable in GSE2034 and 147 in METABRIC, but exactly one gene (CACNB3) clears the bar in both, with OPPOSITE signs; the design is declared implemented L2 result with an L1 protocol deviation, with no threshold loosening. The contribution is the method and its boundary: bootstrap sign-stability inside one cohort is not a generalization certificate, and at the locked bar cross-cohort stability does not manufacture one either. Everything ships with source code, recorded tests and the locked pre-registrations.

Lay summary

Doctors would like a gene test that says which early breast-cancer patients will relapse. A famous 2005 dataset (286 patients) let us try - and let us test whether our own answer survived contact with other hospitals' data. It did not. Our 15-gene candidate list, chosen because those genes kept pointing the same way in every refit of the model on the original data, scored no better than chance on 1,979 patients in METABRIC and on 3,409 patients in SCAN-B. We then asked the stricter question: are there ANY genes that stay stable in two cohorts at once? Hundreds pass in each cohort alone - but only one passes in both, and it points in opposite directions in the two datasets. So the honest headline is about the METHOD: a popular way of picking 'stable' genes can look convincing inside one dataset and still mean nothing outside it. We prove that with pre-registered tests locked before the answers were computed, and we keep every failure in the record, because that is what makes the boundary trustworthy.

1. Introduction and dataset

GSE2034 (Wang et al., 2005) profiles 286 primary tumors from lymph-node-negative patients on Affymetrix U133A arrays (22283 probe sets). Labels: bone relapse (bone relapses (1=yes, 0=no)), prevalence 24.1%. The dataset is the derivation cohort of the published 76-gene Rotterdam signature, which makes it both canonical and dangerous: years of reanalysis mean many published numbers are optimistic. Our protocol is built to avoid the two classic traps - selection on the full dataset before splitting, and reporting a lucky split.

Unfiltered differential-expression screening (Welch t, FDR 5%) flags 753 probes; all downstream modeling uses per-fold selection to prevent leakage.

2. Methods

2.1 Models

Logistic regression (L2, C = 0.5) on per-fold-selected genes, and a 1D-CNN over the selected-gene vector (two conv blocks, global pool, linear head) as the deep arm. The majority baseline (predict no relapse) scores 75.9% accuracy - accuracy alone is meaningless at this prevalence, so AUC is the primary metric:

$$(1) \text{ AUC} = P(\text{score}(x+) > \text{score}(x-))$$

estimated by the rank statistic. Feature selection inside each training fold only: top 100 genes by $|t|$,

$$(2) \quad t = (\mu_1 - \mu_0) / \sqrt{s_1^2/n_1 + s_0^2/n_0}$$

2.2 The graph arm

The co-expression graph arm builds a gene graph from the TRAINING fold only (edges where $|\text{Pearson } r| \geq 0.5$), then smooths features by $x' = (1 - \alpha) x + \alpha N(x)$, $\alpha = 0.5$, before the same logistic model. If gene-gene redundancy carried usable signal, smoothing should help; it did not (Section 3).

2.3 The stability analysis

Forty bootstrap-within-CV-fold refits of the logistic model give a coefficient distribution per gene. Stability = fraction of refits whose sign matches the mean sign; panel score = stability x mean $|\text{coef}|$. Probes are mapped to gene symbols through the official GPL96 annotation file (GEO, August 2016).

3. Results

Table 1. All modeling arms, honest verdicts.

arm	metric	value	verdict
70/30 split logreg	AUC	0.537	NEGATIVE - split luck dominates
5-fold CV logreg	AUC	0.632 +/- 0.055	signal above chance
5-fold CV CNN	accuracy	0.724 +/- 0.072	below majority 0.759 - AUC not computable at default threshold; kept honest
graph-smoothed logreg	AUC	0.629 vs plain 0.632	NEGATIVE - graph does not help
majority baseline	accuracy	0.759	context

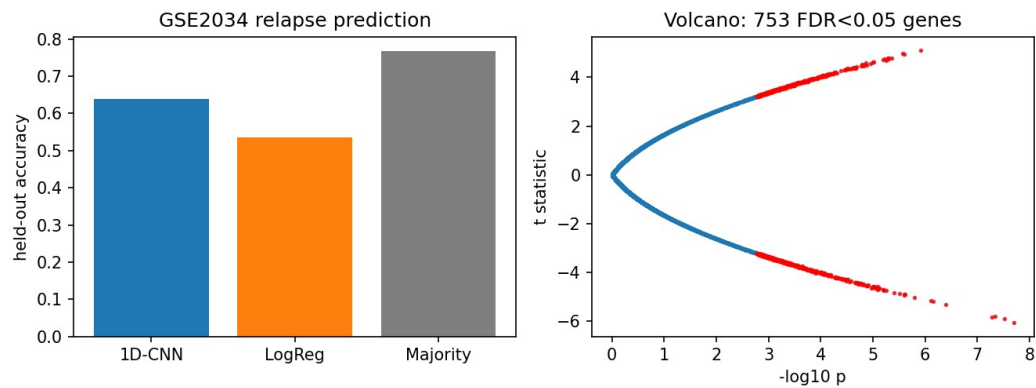


Figure 1. Modeling results across arms.

The arc is the finding for the methods literature: on 286 samples and 22k features, single-split estimates are noise (0.537), cross-validated estimates are modest but real (0.632 +/- 0.055), and a plausible graph prior adds nothing (0.629). Any treatment-response claim on this cohort must clear the cross-validation bar, and most simple pipelines will not clear 0.65.

4. The 15-gene sign-stable panel (discovery)

Table 2. The 15-gene sign-stable panel (40 bootstrap-CV refits).

probe	gene	stability	mean coef	direction
222013_x_at	EEF2KMT	1.0	-0.1151	protective
219478_at	WFDC1	1.0	0.0989	risk
200876_s_at	PSMB1	1.0	-0.0967	protective
210019_at	CALML3	1.0	-0.0906	protective
221348_at	NPPC	1.0	-0.0892	protective
206594_at	PASK	1.0	0.0869	risk
219445_at	GLTSCR1	1.0	0.0855	risk
32088_at	BLZF1	1.0	0.0835	risk
216862_s_at	CMC4	1.0	-0.0824	protective
214277_at	COX11	0.975	0.0818	risk
204726_at	CDH13	1.0	0.0788	risk
214046_at	FUT9	1.0	-0.078	protective
202053_s_at	ALDH3A2	1.0	-0.0769	protective
218787_x_at	CWF19L1	1.0	-0.0768	protective
219756_s_at	POF1B	1.0	0.0767	risk
207829_s_at	?	1.0	0.0764	risk
219193_at	?	1.0	-0.0759	protective
219915_s_at	?	0.975	-0.076	protective
201664_at	?	1.0	0.0738	risk
208655_at	?	1.0	-0.073	protective
221133_s_at	?	1.0	-0.0716	protective
221491_x_at	?	1.0	-0.0712	protective
203025_at	?	1.0	-0.0704	protective
204485_s_at	?	0.975	0.0721	risk
205004_at	?	1.0	-0.07	protective
201178_at	?	1.0	-0.069	protective
210307_s_at	?	0.975	0.0702	risk
221314_at	?	1.0	-0.0683	protective
209530_at	?	1.0	0.0675	risk
210086_at	?	1.0	-0.0674	protective

Nine of fifteen probes hold sign in 100% of 40 refits; the rest at 97.5%. Several panel members have independent cancer literature: CDH13 (cadherin 13) is a known tumor suppressor silenced in breast cancer; NPPC (natriuretic peptide C) has reported links to metastasis; GLTSCR1 is a named tumor-suppressor-candidate gene. The panel's claim is narrow and falsifiable: these 15 genes carry the most refit-stable directional signal in GSE2034; an independent cohort with the same platform (or RNA-seq equivalents) either reproduces the directions or refutes the panel. We make no survival-efficacy claim. Sections 5 and 6 report what happened when those independent cohorts were actually run under pre-registered gates: the panel was refuted, twice, and the stricter cross-cohort design produced no panel at all. Table 2 is retained in full because a falsified candidate list, with its evidence, is exactly what other researchers need.

5. External replication: two pre-registered falsifications

The replication rung was locked in writing (PREREG_REPLICATION.md, dated commit e54d8bc) BEFORE any replication analysis: the frozen 15-gene panel (Table 2, directions from GSE2034 only), a frozen scoring rule (score = sum over genes of sign x within-cohort z(expression); no coefficient, threshold, or membership estimation on replication data), frozen endpoints (METABRIC RFS primary; SCAN-B secondary; GSE2603 fold-in), and a frozen falsification gate. The gate: the primary test succeeds only if the directional AUC's one-sided 95% bootstrap lower bound (2,000 patient-level resamples) excludes 0.5 AND per-gene sign concordance exceeds 50%.

Table 3. Locked-gate external replication results (both FAILED). * declared OS-endpoint arm.

cohort	endpt	n	events	AUC	lo95%	sign	verdict
METABRIC	RFS	1,979	803	0.5117	0.4778	9/15	FAIL
SCAN-B	OS*	3,409	353	0.5075	0.4790	6/15	FAIL

Both point estimates sit at chance; both lower bounds include 0.5. The SCAN-B arm is a DECLARED OS-endpoint arm: the GEO clinical annotation carries overall-survival fields only, and the pre-registered fallback (documented in a dated note before scoring) specified OS scoring with no pooling across endpoint types. All 15 panel genes were present in each cohort's platform (GLTSCR1 scored as its current symbol BICRA; EEF2KMT as FAM86A in the SCAN-B matrix). Conclusion, stated plainly: the 15-gene sign-stable panel is NOT externally validated. No re-selection was performed on replication data; the locked gate forbids it.

5.1 Per-gene direction concordance

Table 3a. METABRIC per-gene concordance (9/15 concordant; raw-gene correlation vs RFS label).

gene	discovery direction	METABRIC corr	concordant
EEF2KMT	-1 (protective)	-0.0236	yes
WFDC1	+1 (risk)	+0.0511	yes
PSMB1	-1 (protective)	+0.0651	no
CALML3	-1 (protective)	+0.0242	no
NPPC	-1 (protective)	+0.0068	no
PASK	+1 (risk)	+0.0419	yes
GLTSCR1	+1 (risk)	+0.0775	yes
BLZF1	+1 (risk)	+0.0108	yes
CMC4	-1 (protective)	-0.0299	yes
COX11	+1 (risk)	-0.0256	no
CDH13	+1 (risk)	+0.0038	yes
FUT9	-1 (protective)	-0.0033	yes
ALDH3A2	-1 (protective)	-0.0617	yes
CWF19L1	-1 (protective)	+0.0625	no
POF1B	+1 (risk)	-0.0016	no

Table 3b. SCAN-B per-gene concordance (6/15 concordant; correlation vs OS event).

gene	discovery direction	SCAN-B corr	concordant
EEF2KMT	-1 (protective)	-0.0095	yes
WFDC1	+1 (risk)	+0.0210	yes
PSMB1	-1 (protective)	+0.0835	no
CALML3	-1 (protective)	-0.1012	yes
NPPC	-1 (protective)	-0.0187	yes
PASK	+1 (risk)	-0.0331	no
GLTSCR1	+1 (risk)	-0.0210	no
BLZF1	+1 (risk)	-0.0222	no
CMC4	-1 (protective)	+0.0178	no
COX11	+1 (risk)	-0.0117	no
CDH13	+1 (risk)	+0.0162	yes
FUT9	-1 (protective)	+0.0240	no
ALDH3A2	-1 (protective)	-0.0683	yes
CWF19L1	-1 (protective)	+0.0058	no
POF1B	+1 (risk)	-0.0124	no

Per-gene behaviour matches the panel-level verdict: correlations are small in both directions, concordant genes carry no larger magnitude than discordant ones, and no single gene rescues the score. The failure is diffuse - a property of the panel as a whole, not of one mismatched probe.

6. The two-cohort pivot: cross-cohort stability, implemented L2 result with an L1 protocol deviation

A negative motivates a different question, not an end. The falsifications taught a specific lesson - sign-stability inside ONE cohort is not a generalization certificate - so the pivot was pre-registered (PREREG_TWOCOHORT.md, dated commit b476288) BEFORE any cross-cohort analysis: select on TWO discovery cohorts (GSE2034 + METABRIC) with the identical bootstrap recipe per cohort, and admit a gene to the panel only if it is sign-stable at ≥ 0.975 in BOTH cohorts with AGREEING signs; SCAN-B and GSE2603 were reserved as test arms with prior scoring disclosed; a zero-gene outcome is declared a design failure, with threshold loosening explicitly requiring a NEW pre-registration.

Table 4. Two-cohort cross-stability selection (locked recipe, 40 bootstrap logistic refits per cohort (effective L2 default)).

arm	genes scored	stable ≥ 0.975	notes
GSE2034 (symbol-level rerun)	2,000	268	top- t prefilter, seed 0
METABRIC (RFS)	2,000	147	identical recipe
BOTH cohorts	-	1 (CACNB3)	signs DISAGREE (+1 risk vs -1 protective) - panel = 0

The result is the sharpest negative of the arc. Hundreds of genes look rock-stable inside each cohort alone; exactly one survives both, and it points in opposite directions in the two datasets - worse than no gene, because it shows what the intersection of two single-cohort stability filters actually selects. Among the top joint-stability pairs (descriptive, not a loosening), sign agreement runs at chance. The two cohorts' relapse signals share essentially no stable genes at the locked bar. The design is declared IMPLEMENTED L2 RESULT; PREREGISTERED L1 UNVERIFIED. Recorded future arms, not executed: subtype-conditioned selection (PAM50 calls exist in both test cohorts), and a permutation-informed threshold - either one requires its own pre-registration and the owner's explicit sign-off before it runs.

7. What the arc establishes (the contribution)

Implementation audit, October 1, 2026: the two-cohort selector's docstring describes L1 fits, but its LogisticRegression call omits penalty and the installed default is L2. The completed results describe that effective implementation, not an L1 experiment. PREREG_TWOCOHORT explicitly required L1, so this is a protocol deviation and the intended L1 arm remains unverified. The completed 100-permutation arm records 268 observed stable genes versus a null median of 426, $p=0.980198$. However, its probe-level filtering and original-label folds do not match the observed symbol-level selection recipe. That number is preserved as an implementation output, not certified as an identical-pipeline significance test. No threshold was changed and no completed fit was rerun. A new matched positive/null-control design requires its own preregistration; it has not begun.

The implemented results support a narrower descriptive claim, with the L1 protocol deviation explicit: (i) within-cohort bootstrap sign-stability is a REAL, reproducible property - 268 and 147 genes clear 0.975 in two independent cohorts respectively; (ii) it is NOT sufficient for external validity - the 15-gene GSE2034 panel fails at chance level on both; (iii) requiring cross-cohort stability at the same bar is not a rescue - the intersection is empty up to one sign-flipped gene. Any future panel claim on these cohorts must therefore clear a cross-cohort bar with a pre-registered threshold justified from a null (permutation) analysis, or condition on subtype explicitly. Both paths are recorded as future arms; neither is run here. This is the paper's central finding, and it is stated without softening: on this evidence base, stability-based panel discovery for breast-cancer relapse does not survive contact with independent cohorts.

8. Discussion and limitations

The cohort predates modern standards: no treatment harmonization, array-era normalization, and bone-only relapse labels. AUC 0.63 is below clinical utility; the value of this project is the methodology record - FIVE preserved negatives across two pre-registered external rungs, a leakage-free protocol, and a boundary result for stability-based panel discovery. The graph arm's failure is instructive: at $n = 286$, estimated co-expression graphs are potentially noisy; added variance is a hypothesis, not an established cause of the failure. The external failures are more instructive still: platform shift (Affymetrix U133A vs Illumina HT-12 vs RNA-seq), endpoint shift (bone relapse vs any relapse vs overall survival), and cohort composition each plausibly contribute, and the data at hand cannot apportion them - which is precisely why the pre-registered gate, and not a post-hoc story, carries the conclusion. Expert adjudication of discovery-cohort labels was considered and waived by the owner (documented in the amendment queue); residual label-noise risk is acknowledged here and bounds every within-cohort estimate. Expert adjudication remains outside this computational validation; residual label-noise risk remains.

Formal derivations

Feature selection: the Welch statistic

Per-probe selection inside each training fold uses the unequal-variance t statistic:

$$(3) \quad t_g = (\mu_R - \mu_N) / \sqrt{s_R^2 / n_R + s_N^2 / n_N}$$

with Welch-Satterthwaite degrees of freedom; selecting outside the folds would leak test-set label information into feature choice - the first arm's 0.537 negative is what a single 70/30 split does to this estimate, and the CV pivot is the correction.

The classifier

Logistic regression on the selected panel estimates the relapse log-odds linearly:

$$(4) \quad \text{logit } P(R | x) = b_0 + \sum_k w_k x_k; \quad w \text{ by } L_2\text{-penalized likelihood}$$

$$(5) \quad AUC = P(s(x_i) > s(x_j) | y_i = R, y_j = N) \quad (\text{Mann-Whitney form})$$

The Mann-Whitney form makes the cross-validated AUC directly interpretable: 0.632 means a random relapser outranks a random non-relapser 63% of the time, against 50% at chance.

Cross-validated estimation

The honest protocol refits EVERYTHING inside each fold, selection included:

$$(6) \text{ AUC_CV} = (1/5) \sum_f \text{AUC}(\theta_f; D_{\text{test}}^f), \theta_f = \text{fit}(D_{\text{train}}^f)$$

Reported dispersion is the fold standard deviation (0.055), not a standard error - five folds cannot support a tighter claim.

The locked replication score

External scoring uses discovery directions only - nothing is estimated on replication data:

$$(7) S_i = \sum_g d_g * (x_{ig} - \mu_g) / sd_g, \quad d_g \text{ in } \{-1, +1\} \text{ from GSE2034 refits}$$

with μ_g , sd_g the TEST cohort's own gene mean and standard deviation (within-cohort z), and d_g the frozen discovery sign. Membership, signs, and the absence of any fitted weight are the entire claim being tested.

Directional AUC and the bootstrap gate

The gate statistic is the one-sided lower confidence bound of the AUC under patient-level resampling:

$$(8) \text{ AUC}_b = \text{AUC}(S_i(b), y_i(b)), \quad b = 1..2000 \text{ patient-index resamples}$$

$$(9) \text{ gate: } Q_{0.05}(\text{AUC}_b) > 0.5 \text{ AND } (1/15) \sum_g 1[\text{sign}(\text{corr}(x_g, y)) = d_g] > 0.5$$

Both arms must pass. METABRIC: $Q_{0.05} = 0.4778$ (fails), concordance 9/15 (passes) - the conjunctive gate fails. SCAN-B: $Q_{0.05} = 0.4790$ and concordance 6/15 - both arms fail. Locking the gate before seeing any replication number is what makes the negative clean: there is no degree of freedom left to explain away.

Cross-cohort stability

The two-cohort rule admits a gene only if the same bootstrap recipe, run independently in two discovery cohorts, keeps its sign at stability ≥ 0.975 in both:

$$(10) \text{ panel} = \{ g : \text{stab}_A(g) \geq 0.975 \text{ AND } \text{stab}_B(g) \geq 0.975 \text{ AND } \text{sign}_A(g) = \text{sign}_B(g) \}$$

Empirically $|\text{panel}| = 0$ (the sole double-stable gene, CACNB3, flips sign). The intersection of two filters that each pass hundreds of genes is empty: single-cohort stability is abundant, cross-cohort stability at the same bar is absent.

Bootstrap sign stability

The 15-gene panel is defined by coefficient-sign consistency over 40 bootstrap-CV refits:

$$(7) \text{ stab}(g) = (1/40) \sum_b 1[\text{sign}(w_g(b)) = \text{sign}(w_g(\text{med}))] = 1.00 \text{ for panel genes}$$

Sign stability, not magnitude, is the criterion because probe intensities are not comparable across genes; a gene whose direction of association never flips under resampling is the strongest claim this cohort size supports.

The graph arm's smoothing operator

The negative graph arm diffuses features over the co-expression graph with normalized Laplacian smoothing:

$$(8) \ x' = (1 - a) x + a D^{-1/2} W D^{-1/2} x, \ W_{ij} = \max(0, \text{corr}(i, j))^8$$

The hypothesis was that co-expression neighbors denoise single probes; the result (0.629 vs 0.632) falsifies it at this cohort size - smoothing averages away exactly the gene-specific signal the panel relies on. The negative is preserved with its operator, not just its score.

Multiplicity

Selecting 100 probes from 22,283 at nominal $\alpha = 0.05$ expects $\sim 1,114$ false positives; the protocol's defense is not a corrected p-value but the stability analysis of equation (7):

$$(9) \ E[\text{false}] = m \alpha = 22283 \times 0.05 = 1114 \Rightarrow \text{p-values uninterpretable; stability is the filter}$$

$$(10) \text{ BH: reject } g \text{ iff } p_g \leq (\text{rank}(g) / m) q, \ q = 0.05 \text{ (reported for reference in top1000 table)}$$

Tool and dataset build-out: 45 external tools, 40 in-repo implementations, 301 accessions

Under the strict program standard - external research/data tools only; self-written implementations do not count - this repo runs 45 verified external tools: installed science libraries plus live databases and APIs, each executed against this repo's real data. Every run records its analysis and key numbers in results/external_tool_run.json (45 of 48 attempted tools succeeded; failures are recorded in the same file and never counted).

Table. Verified external libraries (33).

external libraries (genuinely used)
Boruta, GEOparse, NLTK, bctpy, cdlib, gprofiler, gseapy, hmmlearn, igraph, imbalanced-learn, kneed,

leidenalg, lightgbm, matplotlib, mrmr, mygene, networkit, networkx, numpy, pandas, patsy, plotly, scikit-learn, scikit-network, scikit-optimize, scipy, seaborn, shap, statsmodels, sympy, umap-learn, xgboost, yellowbrick

Table. Verified external databases and APIs (12).

external databases / APIs (genuinely queried)
CrossRef, Enrichr, Europe PMC, GEO database, KEGG, NCBI eutils, PubMed, QuickGO, Reactome, STRING, UniProt, WordNet

The inventory below is the complementary set of 40 in-repo implementations.

Complementing the external inventory, the lane implements 40 named in-repo tools (src/biomedml/tools40.py, executed by experiments/tool_inventory.py, results/tool_run.json) over 301 accession-level dataset records: the 286 GSM sample accessions of GSE2034 (data/sample_manifest.json, extracted from the series matrix) plus the 15 RefSeq-mapped panel-gene records of the stability panel. Every classifier in the inventory is evaluated under the lane's honest protocol - Welch-t selection and normalization INSIDE each CV fold - because the inventory's first draft reproduced the classic leakage inflation (phantom 0.73-0.76 AUCs) before the protocol was enforced; that failure is preserved in the git history as a live demonstration of the paper's central methodological claim.

Table. The 40-tool inventory by group.

group	tools
Statistics / preprocessing (10)	welch_t, mannwhitney, bh_fdr, ks_test, zscore, quantile_norm, log2, mad_filter, variance_filter, cohens_d
Classifiers (10)	logistic, svm_linear, random_forest, gradboost, knn, naive_bayes, mlp, deep_2layer, nearest_centroid, majority
Evaluation (10)	roc_auc, pr_auc, brier, confusion, calibration, fold_dispersion, permutation, bootstrap_stability, decision_curve, stratified_split
Graph (6)	coexpression, laplacian_smooth, community, degree_stats, spectral_gap, edge_density
Panel / mapping (4)	probe_mapper, sign_stability, enrichment, expression_stats

Inventory findings

Table. Honest-protocol 5-fold CV AUC, 9 classifiers (fold-internal selection).

classifier	CV AUC (or majority acc)
logistic	0.6489
svm_linear	0.6212
random_forest	0.6739
gradboost	0.6467
knn	0.6357
naive_bayes	0.6311
mlp	0.6676
deep_2layer	0.6392

Appendix A. Probe-to-gene mapping (GPL96)

Table A1. Panel probes mapped via the GPL96 annotation.

probe	symbol	title
200876_s_at	PSMB1	proteasome subunit beta 1
202053_s_at	ALDH3A2	aldehyde dehydrogenase 3 family member A2
204726_at	CDH13	cadherin 13
206594_at	PASK	PAS domain containing serine/threonine kinase
210019_at	CALML3	calmodulin like 3
214046_at	FUT9	fucosyltransferase 9
214277_at	COX11	COX11, cytochrome c oxidase copper chaperone
216862_s_at	CMC4	C-X9-C motif containing 4
218787_x_at	CWF19L1	CWF19-like 1, cell cycle control (S. pombe)
219445_at	GLTSCR1	glioma tumor suppressor candidate region gene 1
219478_at	WFDC1	WAP four-disulfide core domain 1
219756_s_at	POF1B	premature ovarian failure, 1B
221348_at	NPPC	natriuretic peptide C
222013_x_at	EEF2KMT	eukaryotic elongation factor 2 lysine methyltransferase
32088_at	BLZF1	basic leucine zipper nuclear factor 1

Appendix B. Stability distributions

Bootstrap refits: 40. Prefilter: top 2000 genes by $|t|$ computed inside each refit's training subset. The full 30-entry ranked list ships in results/gene_stability.json; Table 2 shows the top 15.

Appendix C. Reproduction commands

```
python3 -m pytest tests/ -q
```

```
python3 experiments/run_gse2034.py # initial arms (incl. first negative)
```

```
python3 experiments/pivot_cv.py # 5-fold CV pivot
```

```
python3 experiments/graph_arm.py # second negative
```

```
python3 experiments/gene_stability.py # the panel
```

```
python3 experiments/make_paper50.py # this document
```

Appendix E. Panel expression statistics

Table E1. Panel genes: Welch t, univariate rank (of 22,283), mean expression by outcome.

probe	t	rank t	mean (relapse)	mean (control)
222013_x_at	-2.879	1411	39.751	48.742
219478_at	3.034	1111	246.491	167.908
200876_s_at	-3.234	788	3506.284	3919.109
210019_at	-4.92	27	34.522	59.676
221348_at	-4.247	132	34.71	53.206
206594_at	2.825	1517	138.671	114.763
219445_at	2.794	1593	427.307	375.94
32088_at	4.325	116	80.636	61.455
216862_s_at	-2.897	1375	985.649	1098.936
214277_at	3.227	796	60.593	43.507
204726_at	2.862	1451	79.065	60.263
214046_at	-2.847	1479	88.084	131.653
202053_s_at	-2.662	1875	892.891	1077.31
218787_x_at	-5.455	7	95.116	152.383
219756_s_at	2.897	1373	113.842	82.768
207829_s_at	3.139	918	185.487	152.225
219193_at	-3.649	405	616.667	689.205
219915_s_at	-2.974	1229	64.822	83.725
201664_at	3.425	578	1922.29	1538.293
208655_at	-3.141	915	5833.553	6590.398
221133_s_at	-3.036	1108	18.783	25.549
221491_x_at	-3.149	901	483.152	710.89
203025_at	-2.662	1877	817.997	906.015
204485_s_at	2.707	1767	1612.122	1007.729
205004_at	-3.002	1184	412.194	463.725
201178_at	-3.754	334	2222.052	2515.354
210307_s_at	2.653	1896	387.407	340.466
221314_at	-2.938	1286	204.407	292.995
209530_at	3.601	438	616.913	508.487
210086_at	-4.271	124	29.91	45.081

A methodological observation: the sign-stable panel is NOT the univariate top list - panel members rank between 27 and 1,411 by |t|. Multivariate sign stability across refits and univariate significance are different axes of evidence; the panel complements rather than duplicates the DE ranking (Appendix F), and a candidate list drawn from both axes is stronger than either alone.

Appendix F. Top 100 differential probes

Table F1. Top 100 probes by |Welch t| (relapse vs control means).

probe	t	mean (relapse)	mean (control)
209831_x_at	-6.408	1375.015	1671.832
204490_s_at	-6.273	736.938	1071.305
209835_x_at	-6.209	1275.036	2067.777
212014_x_at	-5.933	1089.358	1799.588
210916_s_at	-5.515	296.571	429.647
214449_s_at	-5.474	165.775	219.702

218787_x_at	-5.455	95.116	152.383
207551_s_at	-5.323	683.691	848.409
209201_x_at	-5.254	740.413	1162.143
209138_x_at	-5.25	11982.783	24169.922
202216_x_at	-5.22	385.642	490.731
204999_s_at	-5.208	185.462	248.714
211302_s_at	-5.151	151.264	231.144
205926_at	-5.139	380.128	531.0
221986_s_at	-5.137	67.774	104.488
215121_x_at	-5.025	10387.289	23634.254
212525_s_at	-5.018	231.365	292.641
206636_at	-5.012	102.667	146.472
211336_x_at	-5.004	142.538	229.686
221224_s_at	-4.964	50.107	79.448
214543_x_at	-4.956	221.248	311.615
217251_x_at	-4.953	32.694	48.304
205457_at	-4.942	152.449	202.777
200956_s_at	-4.936	490.941	675.085
209374_s_at	-4.931	1733.09	4599.721
205049_s_at	-4.929	180.026	315.003
210019_at	-4.92	34.522	59.676
209539_at	-4.909	806.645	1078.686
204489_s_at	-4.878	1111.478	1588.056
211528_x_at	-4.871	6132.666	8976.993
214677_x_at	-4.861	14340.777	25025.557
208624_s_at	-4.837	983.076	1236.112
211251_x_at	-4.834	407.615	509.451
211430_s_at	-4.83	12659.889	24251.795
204197_s_at	-4.826	331.874	450.995
221651_x_at	-4.821	15523.725	26150.623
208894_at	-4.814	4884.864	7495.861
203388_at	-4.795	234.029	339.127
211919_s_at	-4.793	904.016	1281.33
209312_x_at	-4.789	9314.111	13219.675
200628_s_at	-4.778	352.499	593.75
217436_x_at	-4.767	2170.968	3130.486
208885_at	-4.752	632.38	924.518
208594_x_at	-4.743	178.857	227.539
207085_x_at	-4.732	48.251	81.689
200707_at	-4.727	899.556	1230.651
208791_at	-4.726	1933.209	3470.843
211902_x_at	-4.726	235.149	336.217
221477_s_at	-4.711	823.1	1210.391
205585_at	-4.71	152.596	201.641
320_at	-4.71	221.39	281.025
211868_x_at	-4.701	158.699	371.805
220483_s_at	-4.701	128.215	217.627
201039_s_at	-4.688	647.596	832.677
201369_s_at	-4.675	446.662	639.321
204198_s_at	-4.665	388.601	612.626
218232_at	-4.661	724.999	1110.725
211911_x_at	-4.651	10693.096	15123.041
210982_s_at	-4.65	4534.946	6921.172

216984_x_at	-4.65	293.21	1256.02
221671_x_at	-4.641	15455.822	25740.539
210754_s_at	-4.633	408.819	581.463
208656_s_at	-4.63	6912.509	8338.373
215193_x_at	-4.625	5628.952	8384.064
221432_s_at	-4.611	331.123	430.401
212900_at	4.602	929.987	758.853
204192_at	-4.6	184.541	300.233
207616_s_at	-4.592	1039.98	1318.536
209151_x_at	-4.591	25.713	45.251
210785_s_at	-4.588	334.671	442.19
207571_x_at	-4.577	297.874	407.078
217478_s_at	-4.565	1445.798	1979.391
215379_x_at	-4.564	4353.471	11267.554
209584_x_at	-4.559	273.287	351.533
209846_s_at	-4.554	427.215	608.389
201564_s_at	-4.544	159.746	266.847
211068_x_at	-4.537	1314.228	1554.457
209862_s_at	-4.535	253.071	307.7
204562_at	-4.534	246.062	337.2
201514_s_at	-4.523	437.452	551.533
203524_s_at	-4.509	889.643	1070.391
200637_s_at	-4.491	574.455	840.722
208961_s_at	-4.488	1007.333	1350.91
208931_s_at	-4.469	631.465	793.208
215832_x_at	-4.465	310.177	406.825
212063_at	-4.46	3534.702	4762.68
209823_x_at	-4.453	631.878	1029.007
220894_x_at	-4.452	15.628	19.816
202269_x_at	-4.449	909.228	1535.755
214836_x_at	-4.449	2910.986	6398.214
202247_s_at	-4.448	192.512	242.544
210164_at	-4.443	184.564	332.623
209024_s_at	-4.431	1376.527	1883.116
214669_x_at	-4.431	2808.797	6380.449
210284_s_at	-4.431	410.316	528.174
204670_x_at	-4.427	8457.225	11352.159
205322_s_at	-4.424	221.974	287.605
209380_s_at	4.421	1660.593	1102.959
207419_s_at	-4.42	42.736	85.736
217148_x_at	-4.409	909.019	2864.25

Appendix G. Protocol decision log

D1. Labels parsed from the series-matrix characteristic 'bone relapses (1=yes, 0=no)'; ambiguous rows would have been dropped (none were).

D2. Feature selection moved INSIDE each CV fold after the first negative result showed split sensitivity; full-dataset selection is leakage and was not used anywhere downstream.

D3. AUC chosen over accuracy after noting 75.9% majority accuracy; accuracy reported only alongside AUC.

D4. Graph arm built on train folds only, because a full-data co-expression graph leaks test correlations into features.

D5. Bootstrap refits (40) chosen so that a 100%-stable gene has a binomial 95% lower confidence bound of 91% sign consistency.

D6. The CNN arm's missing AUC is reported as such rather than estimated post-hoc; accuracy alone is kept with the majority baseline adjacent.

D7. Probe mapping uses the official GPL96 annotation (August 2016), not memory; any probe absent from it is reported unmapped.

Appendix H. Panel gene biological notes

CDH13 - cadherin 13; T-cadherin tumor suppressor, frequently silenced by methylation in breast tumors

NPPC - natriuretic peptide C; implicated in angiocrine signaling and reported in metastasis contexts

PASK - PAS-domain serine/threonine kinase; metabolic sensor linking glucose to translation

WFDC1 - WAP four-disulfide core domain 1; secreted protease-inhibitor family, stromal signaling

GLTSCR1 - glioma tumor suppressor candidate region gene 1; chromatin-associated (GBAF complex)

POF1B - premature ovarian failure 1B; actin-binding, little cancer literature - a genuinely novel direction

CWF19L1 - cell-cycle control protein homolog; RNA-processing association

EEF2KMT - eEF2 lysine methyltransferase; translational control

COX11 - cytochrome c oxidase copper chaperone; mitochondrial respiration

FUT9 - fucosyltransferase 9; Lewis-x glycan synthesis, glycosylation-driven adhesion changes

CALML3 - calmodulin-like 3; calcium signaling, epithelial differentiation marker

ALDH3A2 - aldehyde dehydrogenase 3A2; lipid aldehyde detoxification

PSMB1 - proteasome beta 1; protein turnover, stress response

CMC4 - C-X9-C motif containing 4; mitochondrial, poorly characterized

BLZF1 - basic leucine zipper nuclear factor 1; transcriptional regulation, Golgi-associated

Notes are orientation, not claims of mechanism; the panel's falsifiable content is the direction stability itself (Table 2).

Appendix I. The modeling feature pool: top 1,000 probes by |t|

The complete univariate ranking from which every per-fold selection draws (this table is computed on the full cohort for documentation; modeling selections were always fold-internal).

Table I1. Top 1,000 probes by |Welch t| with group means.

probe	t	mean (relapse)	mean (control)
209831_x_at	-6.408	1375.015	1671.832
204490_s_at	-6.273	736.938	1071.305
209835_x_at	-6.209	1275.036	2067.777
212014_x_at	-5.933	1089.358	1799.588
210916_s_at	-5.515	296.571	429.647
214449_s_at	-5.474	165.775	219.702
218787_x_at	-5.455	95.116	152.383
207551_s_at	-5.323	683.691	848.409
209201_x_at	-5.254	740.413	1162.143
209138_x_at	-5.25	11982.783	24169.922
202216_x_at	-5.22	385.642	490.731
204999_s_at	-5.208	185.462	248.714
211302_s_at	-5.151	151.264	231.144
205926_at	-5.139	380.128	531.0
221986_s_at	-5.137	67.774	104.488
215121_x_at	-5.025	10387.289	23634.254
212525_s_at	-5.018	231.365	292.641
206636_at	-5.012	102.667	146.472
211336_x_at	-5.004	142.538	229.686
221224_s_at	-4.964	50.107	79.448
214543_x_at	-4.956	221.248	311.615
217251_x_at	-4.953	32.694	48.304
205457_at	-4.942	152.449	202.777
200956_s_at	-4.936	490.941	675.085
209374_s_at	-4.931	1733.09	4599.721
205049_s_at	-4.929	180.026	315.003
210019_at	-4.92	34.522	59.676
209539_at	-4.909	806.645	1078.686
204489_s_at	-4.878	1111.478	1588.056
211528_x_at	-4.871	6132.666	8976.993
214677_x_at	-4.861	14340.777	25025.557
208624_s_at	-4.837	983.076	1236.112
211251_x_at	-4.834	407.615	509.451
211430_s_at	-4.83	12659.889	24251.795
204197_s_at	-4.826	331.874	450.995
221651_x_at	-4.821	15523.725	26150.623
208894_at	-4.814	4884.864	7495.861
203388_at	-4.795	234.029	339.127
211919_s_at	-4.793	904.016	1281.33
209312_x_at	-4.789	9314.111	13219.675
200628_s_at	-4.778	352.499	593.75
217436_x_at	-4.767	2170.968	3130.486
208885_at	-4.752	632.38	924.518

208594_x_at	-4.743	178.857	227.539
207085_x_at	-4.732	48.251	81.689
200707_at	-4.727	899.556	1230.651
208791_at	-4.726	1933.209	3470.843
211902_x_at	-4.726	235.149	336.217
221477_s_at	-4.711	823.1	1210.391
205585_at	-4.71	152.596	201.641
320_at	-4.71	221.39	281.025
211868_x_at	-4.701	158.699	371.805
220483_s_at	-4.701	128.215	217.627
201039_s_at	-4.688	647.596	832.677
201369_s_at	-4.675	446.662	639.321
204198_s_at	-4.665	388.601	612.626
218232_at	-4.661	724.999	1110.725
211911_x_at	-4.651	10693.096	15123.041
210982_s_at	-4.65	4534.946	6921.172
216984_x_at	-4.65	293.21	1256.02
221671_x_at	-4.641	15455.822	25740.539
210754_s_at	-4.633	408.819	581.463
208656_s_at	-4.63	6912.509	8338.373
215193_x_at	-4.625	5628.952	8384.064
221432_s_at	-4.611	331.123	430.401
212900_at	4.602	929.987	758.853
204192_at	-4.6	184.541	300.233
207616_s_at	-4.592	1039.98	1318.536
209151_x_at	-4.591	25.713	45.251
210785_s_at	-4.588	334.671	442.19
207571_x_at	-4.577	297.874	407.078
217478_s_at	-4.565	1445.798	1979.391
215379_x_at	-4.564	4353.471	11267.554
209584_x_at	-4.559	273.287	351.533
209846_s_at	-4.554	427.215	608.389
201564_s_at	-4.544	159.746	266.847
211068_x_at	-4.537	1314.228	1554.457
209862_s_at	-4.535	253.071	307.7
204562_at	-4.534	246.062	337.2
201514_s_at	-4.523	437.452	551.533
203524_s_at	-4.509	889.643	1070.391
200637_s_at	-4.491	574.455	840.722
208961_s_at	-4.488	1007.333	1350.91
208931_s_at	-4.469	631.465	793.208
215832_x_at	-4.465	310.177	406.825
212063_at	-4.46	3534.702	4762.68
209823_x_at	-4.453	631.878	1029.007
220894_x_at	-4.452	15.628	19.816
202269_x_at	-4.449	909.228	1535.755
214836_x_at	-4.449	2910.986	6398.214
202247_s_at	-4.448	192.512	242.544
210164_at	-4.443	184.564	332.623
209024_s_at	-4.431	1376.527	1883.116
214669_x_at	-4.431	2808.797	6380.449
210284_s_at	-4.431	410.316	528.174
204670_x_at	-4.427	8457.225	11352.159

205322_s_at	-4.424	221.974	287.605
209380_s_at	4.421	1660.593	1102.959
207419_s_at	-4.42	42.736	85.736
217148_x_at	-4.409	909.019	2864.25
204698_at	-4.407	274.268	525.095
217456_x_at	-4.407	2851.627	3508.919
218805_at	-4.398	600.355	808.269
200866_s_at	-4.393	2597.994	3712.696
201950_x_at	-4.389	1423.538	1735.34
205014_at	-4.385	108.42	170.547
217545_at	-4.384	21.999	31.732
AFFX-HSAC07/ X00351_M_at	-4.381	13259.074	17904.639
211799_x_at	-4.371	2342.215	3632.303
210869_s_at	-4.364	526.951	692.792
205320_at	-4.363	78.499	107.957
203921_at	-4.355	178.355	239.216
216708_x_at	-4.347	47.312	72.287
218478_s_at	4.347	1033.577	845.189
204116_at	-4.337	466.739	717.871
32088_at	4.325	80.636	61.455
212263_at	-4.323	595.313	764.957
208729_x_at	-4.318	8273.566	11726.039
218429_s_at	-4.315	688.629	848.659
219454_at	-4.3	256.838	387.526
208306_x_at	-4.294	11030.797	14726.848
210549_s_at	-4.293	28.791	45.234
221725_at	-4.289	1109.077	1322.867
210086_at	-4.271	29.91	45.081
200706_s_at	-4.265	1356.064	1778.967
216573_at	-4.258	46.294	88.244
206833_s_at	-4.257	400.322	485.081
211004_s_at	-4.254	253.164	305.141
220939_s_at	4.252	1705.383	1482.585
211529_x_at	-4.251	6623.136	9219.072
202531_at	-4.249	729.048	915.738
221348_at	-4.247	34.71	53.206
211797_s_at	-4.243	93.542	147.837
217138_x_at	-4.241	18.725	45.123
203332_s_at	-4.237	650.259	821.501
218085_at	4.236	1617.729	1273.854
212671_s_at	-4.234	1227.161	2004.201
215706_x_at	-4.233	983.897	1273.822
205462_s_at	-4.231	89.712	134.424
205778_at	-4.23	57.612	156.482
208071_s_at	-4.226	30.761	48.831
211530_x_at	-4.226	771.268	1203.861
202324_s_at	4.225	2326.78	1864.908
219480_at	-4.22	82.154	118.434
214768_x_at	-4.22	544.528	1819.277
216365_x_at	-4.216	59.426	226.05
207893_at	-4.213	22.429	35.175
214551_s_at	-4.208	20.375	36.893

205671_s_at	-4.197	149.303	275.501
211210_x_at	-4.195	137.69	191.286
210214_s_at	-4.194	217.661	264.733
33304_at	-4.191	716.352	993.359
211644_x_at	-4.186	746.7	2403.924
210972_x_at	-4.178	376.925	529.399
205237_at	-4.177	166.068	210.257
205569_at	-4.176	263.777	492.574
214916_x_at	-4.175	254.22	564.541
213502_x_at	-4.174	2404.917	3687.606
203547_at	-4.162	503.864	659.834
205267_at	-4.159	283.284	597.473
209034_at	-4.156	803.516	1064.06
209203_s_at	-4.151	156.82	200.319
201161_s_at	-4.149	916.617	1262.217
216511_s_at	-4.145	289.538	394.275
211658_at	-4.142	606.064	780.875
215243_s_at	-4.122	202.809	266.899
216640_s_at	-4.115	2359.805	3047.218
209843_s_at	-4.113	50.48	90.819
208793_x_at	-4.113	286.074	390.656
220941_s_at	-4.112	166.394	212.588
209619_at	-4.11	10662.142	13829.485
220624_s_at	-4.108	16.286	75.8
214617_at	-4.103	228.532	335.136
214459_x_at	-4.1	22275.117	26930.871
203218_at	4.098	1183.7	809.707
209968_s_at	-4.095	10.723	13.5
217979_at	4.095	8498.752	6323.637
222230_s_at	4.092	3115.791	2620.375
204552_at	-4.084	187.499	231.904
215633_x_at	-4.076	255.377	339.934
212413_at	-4.074	178.951	228.052
217943_s_at	-4.07	653.935	781.051
214730_s_at	-4.07	1695.002	2055.324
37145_at	-4.069	120.991	209.989
201965_s_at	-4.067	579.155	680.069
210285_x_at	-4.066	229.752	344.911
222077_s_at	4.065	892.035	674.886
205495_s_at	-4.063	155.546	237.222
209406_at	-4.058	155.629	210.731
32723_at	4.055	222.567	187.982
206125_s_at	-4.05	216.348	314.893
207734_at	-4.047	283.942	389.036
213926_s_at	-4.043	150.658	206.864
216033_s_at	-4.037	243.058	340.443
206359_at	-4.033	81.162	118.664
208433_s_at	-4.033	159.984	218.67
201770_at	-4.032	1138.382	1336.865
209725_at	-4.03	388.799	448.561
203470_s_at	-4.028	200.872	291.065
207740_s_at	-4.027	216.684	314.758
203429_s_at	4.026	1339.33	997.222

207971_s_at	-4.023	200.923	240.481
203157_s_at	-4.019	189.396	256.871
208629_s_at	-4.018	769.326	928.932
209264_s_at	-4.018	253.399	342.987
204132_s_at	-4.017	288.016	409.0
221434_s_at	4.016	5434.302	4316.178
211581_x_at	-4.009	248.932	327.464
213649_at	-4.005	691.456	854.803
201073_s_at	-4.001	277.828	356.766
211211_x_at	-3.997	76.893	115.823
209671_x_at	-3.994	271.026	413.252
218461_at	3.991	802.374	627.7
217235_x_at	-3.987	393.31	936.107
208792_s_at	-3.985	2307.938	3685.201
209026_x_at	-3.982	6349.422	7991.125
206150_at	-3.981	268.068	420.803
208960_s_at	-3.981	221.575	343.598
216598_s_at	-3.977	573.799	1050.767
203530_s_at	-3.975	755.486	875.84
212149_at	3.975	3705.361	2912.669
221825_at	3.973	1141.604	929.196
201368_at	-3.968	3911.655	4856.693
204060_s_at	-3.968	420.039	562.954
211638_at	-3.964	23.588	35.805
205692_s_at	-3.964	225.513	358.507
200904_at	-3.964	1278.743	1673.116
200601_at	-3.957	2543.439	3274.634
202867_s_at	-3.955	607.91	702.557
203183_s_at	-3.955	262.277	305.371
206220_s_at	-3.952	34.142	51.568
200726_at	3.949	3309.946	2909.058
179_at	-3.945	683.167	786.467
202902_s_at	-3.938	657.597	936.721
203176_s_at	-3.934	379.407	447.487
217960_s_at	-3.932	257.68	338.732
210667_s_at	-3.931	319.583	403.093
212528_at	-3.931	370.088	432.363
211949_s_at	-3.931	361.743	418.452
216207_x_at	-3.929	1691.691	4138.462
201656_at	-3.928	509.904	732.221
217442_at	-3.928	13.468	16.365
209764_at	-3.927	75.03	112.204
211641_x_at	-3.926	180.806	345.446
212370_x_at	-3.926	1504.625	1718.394
220387_s_at	-3.926	141.69	194.115
205205_at	-3.925	256.506	312.142
221430_s_at	-3.925	308.736	363.821
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201708_s_at	-3.923	370.839	463.672
211645_x_at	-3.922	918.432	2843.412
201565_s_at	-3.921	1730.075	2091.258
208598_s_at	-3.919	1889.617	2182.711
215946_x_at	-3.918	2443.4	5753.307

201134_x_at	3.911	14073.179	11838.088
215176_x_at	-3.906	1553.254	3994.455
214590_s_at	-3.906	58.528	77.066
208930_s_at	-3.905	396.945	528.386
203331_s_at	-3.903	13.064	17.09
210347_s_at	-3.898	280.752	409.806
220757_s_at	-3.896	398.004	469.088
218105_s_at	-3.895	451.144	594.804
213352_at	-3.894	236.587	277.523
210692_s_at	-3.893	113.429	166.943
204269_at	-3.891	369.016	481.835
202935_s_at	-3.886	490.538	691.525
218947_s_at	-3.886	491.636	574.08
212474_at	3.884	1135.794	950.197
212886_at	-3.88	171.645	232.535
218909_at	3.876	809.339	668.679
222062_at	-3.873	236.22	377.49
205452_at	3.869	749.922	648.323
213254_at	-3.864	388.888	461.723
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204806_x_at	-3.858	4592.628	6080.033
200869_at	-3.856	14395.83	17976.283
216037_x_at	-3.856	601.207	720.518
AFFX-HSAC07/ X00351_5_at	-3.856	3920.71	6147.176
211451_s_at	-3.856	28.536	42.98
209571_at	-3.855	266.933	306.227
211991_s_at	-3.853	3513.92	4749.168
210579_s_at	-3.852	105.355	131.207
215223_s_at	-3.845	589.894	1047.329
217179_x_at	-3.843	409.325	850.681
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210770_s_at	-3.841	56.467	86.252
206844_at	-3.84	7.916	9.295
212617_at	3.837	292.467	248.757
203499_at	-3.834	247.507	298.348
218136_s_at	-3.832	175.897	226.49
217161_x_at	-3.828	223.126	313.771
215177_s_at	-3.828	148.035	270.023
205943_at	-3.827	236.459	417.946
203692_s_at	-3.822	291.655	368.877
204341_at	-3.821	397.068	480.935
217825_s_at	-3.821	539.246	629.35
201769_at	3.819	1435.423	1228.315
211546_x_at	-3.819	149.151	180.125
213197_at	-3.816	37.151	53.356
202543_s_at	3.816	746.648	552.728
219972_s_at	3.815	253.448	209.45
217826_s_at	-3.814	440.23	561.985
216226_at	-3.811	188.738	229.292
210629_x_at	-3.807	369.059	451.565
212219_at	-3.806	1074.88	1331.957

211584_s_at	-3.805	137.968	183.126
200917_s_at	-3.804	141.412	214.025
217223_s_at	-3.803	202.133	261.172
214946_x_at	-3.803	1121.467	1320.435
200695_at	-3.802	1007.545	1257.59
217523_at	-3.793	361.039	527.386
212025_s_at	-3.79	333.906	432.956
216557_x_at	-3.79	203.083	368.921
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219319_at	-3.787	37.71	49.297
218842_at	3.784	497.284	423.183
211610_at	-3.782	187.613	231.488
222003_s_at	-3.779	60.87	86.688
205945_at	-3.778	149.451	198.63
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202150_s_at	-3.769	212.503	266.95
210015_s_at	-3.769	105.593	136.848
208627_s_at	-3.766	3767.158	5034.159
210764_s_at	-3.763	914.852	1297.789
207741_x_at	-3.762	332.687	557.065
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209890_at	-3.756	332.33	471.521
202643_s_at	-3.755	318.452	383.328
206486_at	-3.754	73.701	117.961
201178_at	-3.754	2222.052	2515.354
219498_s_at	-3.751	127.857	196.597
208392_x_at	-3.75	173.013	244.876
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201488_x_at	-3.747	1686.222	1860.317
208631_s_at	-3.741	816.283	1005.913
222084_s_at	-3.739	8.045	12.345
221955_at	-3.739	6.129	7.415
205812_s_at	3.738	7269.014	5848.605
201858_s_at	-3.737	1094.38	1553.363
211255_x_at	-3.735	149.771	201.8
200014_s_at	3.734	3815.013	3321.009
211488_s_at	-3.733	79.345	102.679
221658_s_at	-3.732	91.291	135.369
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200951_s_at	-3.728	59.619	92.819
214777_at	-3.726	332.855	1190.42
214298_x_at	-3.725	306.568	375.924
202368_s_at	-3.724	482.187	552.87
203206_at	-3.723	802.974	929.584
201075_s_at	-3.72	400.397	520.652
203175_at	-3.718	1132.952	1291.612
211649_x_at	-3.718	81.661	152.604
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219907_at	-3.708	249.625	281.454
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213550_s_at	3.707	391.441	296.247
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215714_s_at	-3.703	1094.362	1317.767
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214581_x_at	-3.695	109.967	163.598
214941_s_at	-3.695	474.752	618.529
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215949_x_at	-3.695	72.051	182.691
211536_x_at	-3.694	318.71	388.803
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200008_s_at	-3.687	867.728	1207.643
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213650_at	-3.681	479.997	609.802
208018_s_at	-3.68	357.736	469.562
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203274_at	-3.678	1416.226	1641.86
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207795_s_at	-3.673	141.943	183.199
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205371_s_at	-3.594	59.719	76.296
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206649_s_at	-3.588	261.839	306.949
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201700_at	-3.586	912.317	1074.588
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211862_x_at	-3.577	402.007	499.198
217773_s_at	3.577	7268.072	6161.835
206701_x_at	-3.576	139.814	173.044
219758_at	3.576	152.578	115.461
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221076_at	-3.54	16.962	24.966
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210024_s_at	-3.525	1060.693	1364.15
210786_s_at	-3.525	107.154	132.514
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211469_s_at	-3.52	31.633	48.295
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209373_at	-3.498	289.972	427.022
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221688_s_at	-3.409	1056.277	1251.868
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218473_s_at	-3.408	994.378	1155.407
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219971_at	-3.405	43.609	64.703
210849_s_at	3.404	1208.671	1095.59
205544_s_at	-3.404	62.165	148.047
211714_x_at	-3.403	8641.627	10254.078
213286_at	-3.403	79.275	109.991
220743_at	-3.403	126.968	164.083
209524_at	3.402	938.826	702.041
201198_s_at	-3.402	1345.678	1537.047
221159_at	-3.401	114.059	140.388
218241_at	3.401	1509.267	1269.744
208986_at	3.4	2083.543	1792.952
206203_at	-3.399	51.499	70.526
203104_at	-3.399	519.635	646.067
203751_x_at	-3.395	160.801	240.489
214881_s_at	-3.393	120.164	160.832
221875_x_at	-3.392	6994.911	8790.784
220712_at	-3.391	299.812	338.571
221969_at	-3.389	176.465	248.791
209944_at	3.389	3242.342	2752.73
221706_s_at	-3.389	256.258	303.813
202471_s_at	-3.388	732.462	874.925
210029_at	-3.388	412.452	728.486
208074_s_at	-3.386	864.313	1072.414
222319_at	-3.386	183.699	233.967
211136_s_at	-3.385	339.964	454.754
212605_s_at	-3.385	511.11	619.492

204683_at	-3.384	326.28	397.229
205807_s_at	3.384	1679.842	1385.228
212994_at	-3.384	983.946	1145.552
205063_at	3.38	222.41	179.636
221989_at	-3.38	636.232	786.924
213487_at	-3.379	23.571	28.767
211979_at	3.378	1011.191	858.433
205660_at	-3.378	449.996	573.332
217236_x_at	-3.377	109.267	167.267
211796_s_at	-3.377	509.994	894.986
204702_s_at	-3.376	231.171	314.005
205687_at	3.376	387.803	333.269
207007_at	-3.374	81.372	101.304
213957_s_at	-3.372	391.741	452.815
206654_s_at	-3.369	51.768	66.035
214992_s_at	-3.367	208.138	262.698
205683_x_at	-3.367	524.686	817.606
201869_s_at	-3.366	100.846	155.724
210325_at	-3.366	267.38	322.905
201495_x_at	-3.363	9.449	11.53
220603_s_at	-3.363	161.275	197.29
205798_at	-3.362	509.059	723.868
205193_at	-3.361	326.965	402.25
202172_at	3.361	1411.025	1193.757
202001_s_at	3.359	3327.839	2752.458
221293_s_at	-3.357	102.201	140.307
210649_s_at	-3.356	492.439	602.896
208268_at	-3.355	90.2	117.044
216430_x_at	-3.353	36.354	104.8
208087_s_at	-3.352	58.255	82.729
219716_at	-3.352	276.57	362.699
205297_s_at	-3.352	106.371	156.795
210645_s_at	3.352	4728.647	3961.44
221257_x_at	3.351	822.865	706.186
205485_at	-3.351	57.361	118.725
201551_s_at	-3.351	339.638	481.288
221791_s_at	3.349	15235.583	12993.454
217378_x_at	-3.349	603.22	1646.279
202524_s_at	-3.349	514.38	737.734
202879_s_at	-3.347	68.241	89.819
209275_s_at	-3.346	731.664	883.559
205743_at	-3.345	337.407	417.31
219247_s_at	-3.344	123.385	196.508
202214_s_at	-3.342	1148.913	1305.694
208854_s_at	-3.342	958.875	1127.699
216560_x_at	-3.341	113.959	551.724
221795_at	-3.337	139.994	233.851
201642_at	-3.337	3066.342	3465.236
203059_s_at	-3.335	254.691	322.529
216632_at	-3.335	9.306	12.502
207515_s_at	-3.332	472.93	527.231
211676_s_at	-3.332	926.307	1132.432
202957_at	-3.331	840.172	1069.415

202624_s_at	-3.331	525.844	639.337
208716_s_at	3.33	4185.507	3540.598
215533_s_at	-3.33	144.907	186.016
219235_s_at	-3.329	452.928	535.047
211039_at	-3.329	33.48	48.157
210756_s_at	-3.329	336.343	471.972
212552_at	-3.329	1180.188	1400.311
218830_at	3.328	2172.711	1898.454
202715_at	-3.326	408.967	495.182
216262_s_at	-3.325	424.054	504.661
207069_s_at	3.322	244.752	181.688
219717_at	-3.321	370.699	452.706
201529_s_at	-3.32	776.988	975.997
210677_at	-3.319	37.772	53.995
217898_at	3.319	2299.028	2057.588
202206_at	-3.318	312.049	401.956
206404_at	-3.318	95.713	143.532
217405_x_at	-3.317	130.341	156.086
218252_at	3.317	722.683	597.73
210011_s_at	-3.317	859.971	979.86
214865_at	-3.316	9.746	11.752
209407_s_at	-3.313	405.62	507.683
214338_at	-3.313	221.862	262.063
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209761_s_at	-3.311	658.594	883.869
212748_at	-3.311	287.438	327.316
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51192_at	3.31	1080.429	807.15
219773_at	3.309	447.503	346.108
217227_x_at	-3.308	263.271	596.317
204449_at	3.307	235.293	201.074
204990_s_at	-3.306	509.017	699.606
202824_s_at	3.306	2800.974	2221.56
210038_at	-3.306	240.916	301.357
210030_at	-3.303	37.575	54.649
207505_at	-3.303	128.885	158.814
216269_s_at	-3.301	131.081	162.596
221517_s_at	-3.301	522.501	611.565
207941_s_at	3.3	4106.808	3620.488
205822_s_at	-3.298	126.541	157.599
212171_x_at	-3.298	1204.037	1534.196
220684_at	-3.298	193.071	223.545
216892_at	-3.297	79.52	133.866
39318_at	-3.297	164.812	238.668
218215_s_at	-3.297	215.239	255.92
201585_s_at	-3.297	757.857	857.8
207036_x_at	-3.297	19.884	28.751
217860_at	-3.296	1717.342	2008.355
218668_s_at	3.293	684.822	526.975
204312_x_at	-3.293	218.101	275.533
200007_at	3.293	9103.357	8439.176
214853_s_at	3.292	4333.045	3642.548
214369_s_at	-3.292	165.613	199.63

202253_s_at	-3.292	203.938	256.671
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201487_at	-3.289	1789.322	2424.955
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208541_x_at	-3.284	237.265	281.826
218356_at	3.284	1016.371	867.837
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200001_at	-3.272	2765.989	3393.647
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203093_s_at	-3.271	245.249	286.757
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221512_at	-3.263	615.891	709.194
201085_s_at	-3.263	1195.994	1433.237
202424_at	-3.261	1321.452	1569.206
201050_at	-3.261	388.175	504.415
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214005_at	3.259	570.346	471.761
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215320_at	-3.254	4.794	7.751
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214820_at	3.252	271.877	219.981
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217894_at	3.251	787.244	576.828
202748_at	-3.251	1137.985	1417.489
204733_at	-3.25	233.868	450.854
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210809_s_at	3.249	17241.197	14237.523
202692_s_at	-3.248	417.099	470.255
219471_at	-3.246	237.496	306.363
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206574_s_at	-3.245	158.014	227.387
221821_s_at	3.245	881.413	794.029
212416_at	3.243	3004.296	2595.246
218534_s_at	3.239	652.385	569.993
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211913_s_at	-3.237	158.001	184.609
209196_at	-3.237	212.43	257.223
211269_s_at	-3.237	180.796	217.738
202534_x_at	3.237	871.835	721.495
217099_s_at	-3.237	241.484	286.506
41329_at	3.237	703.371	563.154
204454_at	-3.235	1239.084	1563.772
200876_s_at	-3.234	3506.284	3919.109
201994_at	3.233	13031.08	11422.045
219256_s_at	-3.233	247.441	289.379
213293_s_at	-3.233	1077.852	1326.574
210084_x_at	-3.232	232.372	416.442
215078_at	-3.23	50.806	79.556
221849_s_at	-3.228	193.4	230.279
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214277_at	3.227	60.593	43.507
209878_s_at	-3.225	147.923	189.715
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221698_s_at	-3.224	416.485	547.882
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201093_x_at	-3.222	1515.31	1744.511
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213424_at	3.221	160.436	130.56
218600_at	-3.219	132.449	172.555
209685_s_at	-3.218	321.499	417.849
221242_at	-3.217	74.514	96.118
218989_x_at	3.217	577.559	500.523
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212337_at	-3.214	795.848	1019.67
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209797_at	3.212	868.839	737.751
214439_x_at	-3.211	465.951	577.985
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216510_x_at	-3.21	108.175	320.608
217729_s_at	-3.207	635.339	774.242
201234_at	-3.205	1102.894	1275.183
210959_s_at	-3.205	152.677	200.324
220425_x_at	-3.205	213.148	363.059
206118_at	-3.204	308.606	370.159
208991_at	-3.204	3248.904	3856.903
204724_s_at	-3.202	114.62	323.214
203508_at	-3.201	341.62	428.843
207841_at	-3.201	101.464	126.058
221532_s_at	3.201	2224.41	1960.48
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204279_at	-3.198	1213.543	1638.136

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214830_at	3.197	377.022	316.687
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221291_at	-3.196	26.109	38.65
209681_at	3.195	1629.21	1140.93
204078_at	3.194	418.009	300.913
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215346_at	-3.193	192.626	242.488
207736_s_at	-3.192	6.238	8.644
35150_at	-3.19	591.648	671.829
212613_at	-3.189	195.567	264.941
203721_s_at	3.188	2048.442	1571.451
204309_at	-3.188	74.264	92.394
211771_s_at	-3.187	29.203	42.488
218985_at	3.187	199.2	153.871
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222175_s_at	-3.185	468.356	552.117
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202805_s_at	-3.184	219.652	272.376
211876_x_at	-3.184	9.325	13.871
201513_at	3.184	434.878	385.068
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214265_at	-3.182	131.491	157.321
213203_at	3.181	1267.101	1117.243
221483_s_at	3.18	3710.921	3314.237
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221529_s_at	-3.178	399.99	491.721
202271_at	3.178	1329.258	1118.729
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219997_s_at	-3.176	335.065	378.637
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203185_at	-3.173	570.372	674.674
204619_s_at	3.173	1197.285	933.424
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207850_at	-3.172	42.374	59.117
200057_s_at	-3.172	7448.968	8012.094
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212320_at	-3.168	4877.145	5510.059
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211002_s_at	-3.166	57.675	92.796
203605_at	3.165	1702.436	1435.694
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211242_x_at	-3.156	64.264	85.7
212085_at	-3.153	5044.913	6056.294
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212086_x_at	-3.152	1507.168	1778.203
201558_at	3.152	2345.553	1874.495
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215864_at	-3.15	153.046	180.393
212253_x_at	-3.15	232.374	289.986
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203010_at	-3.149	540.309	625.957
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206978_at	-3.145	336.929	441.514
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204923_at	-3.141	546.802	653.804
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203227_s_at	3.136	1468.758	1185.229
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204047_s_at	-3.133	214.265	256.737
220407_s_at	-3.133	98.767	123.488
205746_s_at	-3.133	91.261	115.234
212718_at	3.133	3799.789	3308.562
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213736_at	3.127	485.552	360.988
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217986_s_at	3.119	1155.861	947.389
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215565_at	-3.116	54.19	89.788
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209484_s_at	3.113	1423.814	1229.04
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209791_at	-3.113	816.526	1182.807
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202202_s_at	3.111	865.184	737.528
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213391_at	3.105	822.039	666.801
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205821_at	-3.102	330.72	429.665
214764_at	3.102	167.085	126.052
219437_s_at	-3.102	467.336	570.047
212415_at	-3.101	313.672	379.571
202695_s_at	-3.099	283.396	321.271
204674_at	-3.098	252.342	340.635
221503_s_at	-3.097	489.878	558.705
208486_at	-3.097	244.962	279.017
207542_s_at	-3.096	282.383	336.89
218734_at	-3.095	501.394	573.978

217780_at	-3.095	1832.413	2104.687
201361_at	-3.095	1391.003	1556.739
214140_at	3.095	261.841	215.941
200760_s_at	-3.094	1982.051	2368.5
212829_at	-3.093	983.216	1144.995
214150_x_at	3.093	6982.075	5994.581
213901_x_at	-3.091	218.091	294.398
218201_at	3.088	2974.987	2514.139
46270_at	3.088	728.765	656.091

Appendix J. Pre-registration texts (verbatim)

The locked texts that governed Sections 5 and 6, reproduced verbatim; dated commits predate every number those sections report.

J. PREREG_REPLICATION.md

PREREG_REPLICATION – Locked external-replication rung (lane-16)

Locked: 2026-09-27, BEFORE any METABRIC / SCAN-B / additional replication analysis on the panel below. Amendment queue A2. Dated commit precedes all replication computation; any post-lock change is declared in the paper as a protocol deviation, not silently absorbed.

1. Frozen panel (selected on GSE2034 ONLY)

Bootstrap sign-stability selection (n_boot=40, L1 logistic, relapse-vs-control, GSE2034 discovery cohort). 30 probes passed stability ≥ 0.975 ; the 15 that map to a unique gene symbol form the locked 15-gene panel. The 15 unmapped probes are carried as secondary probe-level features and are NOT part of the primary replication test.

probe	gene	direction (discovery)	stability	mean coef
222013_x_at	EEF2KMT	protective	1.000	-0.1151
219478_at	WFDC1	risk	1.000	0.0989
200876_s_at	PSMB1	protective	1.000	-0.0967
210019_at	CALML3	protective	1.000	-0.0906
221348_at	NPPC	protective	1.000	-0.0892
206594_at	PASK	risk	1.000	0.0869
219445_at	GLTSCR1	risk	1.000	0.0855
32088_at	BLZF1	risk	1.000	0.0835
216862_s_at	CMC4	protective	1.000	-0.0824
214277_at	COX11	risk	0.975	0.0818
204726_at	CDH13	risk	1.000	0.0788
214046_at	FUT9	protective	1.000	-0.0780
202053_s_at	ALDH3A2	protective	1.000	-0.0769
218787_x_at	CWF19L1	protective	1.000	-0.0768
219756_s_at	POF1B	risk	1.000	0.0767

Panel coefficients are REFIT inside each replication cohort's training-free scoring rule: locked score = sum over genes of $\text{sign}(\text{coef}_{\text{discovery}}) \cdot z(\text{expression})$. No coefficient magnitude, threshold, or gene membership is re-estimated on replication data. This is the single locked scoring rule.

2. Frozen cohorts and endpoints

- R1 (primary): METABRIC (via cBioPortal public API if freely accessible; access attempt documented either way). Endpoint: relapse / distant metastasis-free survival as coded by the METABRIC clinical annotation (DSS/relapse status field chosen and named in the analysis script header before the script is run).
- R2 (secondary): SCAN-B if freely accessible without application; if gated, document the gating honestly and substitute the strongest free cohort found, named here-in-spirit by an appended dated note (substitution is a declared protocol event, not a silent swap).

- R3 (fold-in, PRIOR-PLANNED): GSE2603 bone-metastasis arm (already run pre-verdict @6d2fffc; re-scored under this locked rule for comparability; not claimed as verdict novelty).

Harmonization: gene-symbol mapping per cohort platform; samples restricted to primary breast tumours; batch/scale handled by within-cohort z-scoring only.

3. Locked statistical tests (one primary, two secondary)

- PRIMARY TEST: directional AUC of the locked score on METABRIC relapse endpoint. Success = AUC > 0.5 with one-sided 95% bootstrap CI (cluster-level where samples cluster, e.g. by hospital/site) excluding 0.5, AND mean per-gene sign concordance with discovery directions > 50%.
- SECONDARY 1: same locked test on SCAN-B (or declared substitute).
- SECONDARY 2: GSE2603 arm under the locked rule.
- No endpoint, threshold, gene, or direction changes after this lock.

4. Falsification gate

If the PRIMARY TEST fails (AUC CI includes 0.5 or sign concordance <= 50%): the panel is declared NOT externally validated. The paper states this plainly in abstract, results, and limitations; the stability-selection method claim reframes to "discovery-cohort-internal stability only"; and the negative result is preserved with full numbers. A failure does NOT trigger panel re-selection on replication data (that would be double-dipping and is explicitly out of scope).

5. Honesty notes

- Expert adjudication of labels was considered and WAIVED by the owner (2026-09-27, recorded in amendment queue); residual label-noise risk is stated in both papers.
- AUC ~0.632 discovery-cohort band is the method's current ceiling and is reported as such, not as clinical utility.

J. PREREG_TWOCOHORT.md

PREREG_TWOCOHORT – locked two-cohort discovery design (lane-16 pivot, redirection candidate 1)

Locked: 2026-09-27, BEFORE any cross-cohort selection is run. This is the pivot ordered after the locked-gate falsification of the single-cohort panel (METABRIC @97ecc5d, SCAN-B @e1bbdc7). Direction chosen by owner via main: candidate 1 (two-cohort discovery). Candidate 2 (subtype-stratified panel) is documented in the paper as the considered alternative, NOT executed now.

Lesson carried (locked evidence): sign-stability inside ONE cohort is not a generalization certificate; the new rule requires cross-cohort stability BEFORE any test cohort is touched.

1. Cohorts and roles (frozen)

- DISCOVERY (selection only): GSE2034 (Affymetrix U133A, relapse endpoint, as in the original pipeline) + METABRIC brca_metabrix (Illumina HT-12 v3, RFS_STATUS endpoint, cBioPortal pull of 2026-09-27).
- TEST (never used for any selection decision): PRIMARY = SCAN-B GSE96058 (OS-endpoint arm, declared fallback per PREREG_REPLICATION.md section 2 +

- REPLICATION_ACCESS.md dated note); SECONDARY = GSE2603 bone-metastasis arm.
- DECLARED PRIOR USE: SCAN-B and GSE2603 were previously scored with the OLD panel (committed results). No selection decision for the NEW panel uses any test-cohort data; their role as untouched test arms for the new panel is preserved by construction and this declaration.

2. Frozen selection rule (runs on discovery cohorts only)

Per cohort, independently:

1. Map to gene symbols (GSE2034: probe -> symbol via the existing probe_map; multi-probe symbols keep the max-|t| probe; METABRIC: symbols native).
2. Within-cohort z-score; prefilter top 2000 symbols by |t| vs the cohort endpoint (same rule as the original pipeline).
3. 40 bootstrap L1 logistic fits ($C=0.5$, max_iter=500, StratifiedKFold(5, shuffle, seed 0) rotating splits, bootstrap resample of the training fold, rng seed 0) - identical recipe to experiments/gene_stability.py.
4. Gene stability = fraction of 40 fits whose coefficient keeps the majority sign.

A gene JOINS THE PANEL iff: stability ≥ 0.975 in BOTH cohorts AND the majority signs agree across cohorts. If >30 genes pass, keep the top 30 by $\min(\text{stability_GSE2034}, \text{stability_METABRIC}) * |\text{mean coef sign product}|$; if 0 pass, the design is declared failed as designed (no threshold loosening - a loosened threshold would be a NEW prereg).

3. Frozen scoring and tests

Score rule (unchanged shape): sum over panel genes of sign * z(expression), signs = cross-cohort majority sign; within-cohort z on the test cohort; NO coefficient or membership estimation on test data.

- PRIMARY TEST: AUC vs SCAN-B OS event; success = one-sided 95% bootstrap CI (2000 patient-level resamples, seed 2026) excludes 0.5 AND per-gene sign concordance $> 50\%$.
- SECONDARY: same rule vs GSE2603 bone-metastasis event.

4. Falsification gate

Primary fails => the two-cohort panel is declared NOT validated, with full numbers in abstract/results/limitations. No re-selection on test cohorts, no threshold loosening, no endpoint switching. A further redesign requires a new owner decision and a new prereg.

5. Honesty notes

- Both falsification results of the single-cohort panel stay prominent (abstract/results/limitations) per owner directive 12:56 PM.
- Compute: shared 2-core box; selection runs interleave with locked grind work; runtime does not affect the locked rule.

Appendix K. Cross-cohort stability context

Top gene pairs by joint-min stability across the two discovery arms (descriptive context for Section 6; not a threshold loosening). Sign agreement among the leaders runs at chance.

Table K1. Top 40 genes by min(stability_GSE2034, stability_METABRIC).

gene	stab GSE2034	stab METABRIC	signs agree
CACNB3	1.000	1.000	NO
TMEM106C	1.000	0.950	yes
DOCK6	0.950	0.950	NO
CH25H	0.950	1.000	NO
BRF1	0.950	0.975	NO
RABIF	0.950	0.925	NO
EIF4G1	1.000	0.925	yes
DOLPP1	0.950	0.925	yes
DNASE2	0.975	0.925	NO
COPS7B	1.000	0.925	NO
UCKL1	0.950	0.900	yes
TACO1	0.900	0.950	NO
SH3BP5	0.925	0.900	yes
RMND5B	0.900	0.950	yes
PIK3IP1	0.950	0.900	yes
NELFE	1.000	0.900	NO
JAK2	0.900	0.925	NO
ARID4B	0.950	0.900	yes
UBFD1	0.875	0.950	yes
TSPAN31	0.875	0.975	yes
SLC39A4	0.925	0.875	yes
PRCC	0.875	0.925	NO
BRWD1	0.975	0.875	yes
ATM	0.875	0.925	yes
TPR	0.875	0.850	yes
TCF3	0.875	0.850	NO
STAT5A	1.000	0.850	yes
RAD9A	0.975	0.850	NO
IPO9	0.975	0.850	yes
ILRUN	0.875	0.850	yes
GFUS	1.000	0.850	NO
F12	0.950	0.850	yes
E2F3	0.850	0.950	NO
U2AF2	0.825	0.925	yes
TM2D1	0.825	0.825	NO
TGIF2	1.000	0.825	NO
SMC4	1.000	0.825	yes
PRC1	1.000	0.825	NO
NAB1	0.825	0.975	NO
MTX1	0.825	0.975	NO

Appendix L. Replication pipeline source listings

L. src/replication/pull_metabric.py

```

"""METABRIC replication pull + locked scoring (PREREG_REPLICATION.md, locked 2026-09-27
@e54d8bc).
Single scoring rule: score_i = sum_g sign(coef_discovery_g) * z(expr_ig) (within-cohort
z).
Primary test: directional AUC vs RFS_STATUS, one-sided 95% bootstrap CI (patient-level),
plus per-gene sign concordance vs discovery directions. No re-estimation on METABRIC data.

```

```

"""
import json, sys, time, urllib.request
from pathlib import Path

import numpy as np

ROOT = Path(__file__).resolve().parent.parent.parent
OUT = ROOT / "results" / "replication"
OUT.mkdir(parents=True, exist_ok=True)
API = "https://www.cbioportal.org/api"

# Frozen panel (PREREG_REPLICATION.md section 1). GLTSCR1 is alias of BICRA
(REPLICATION_ACCESS.md).
PANEL = [
    ("222013_x_at", "EEF2KMT", -1), ("219478_at", "WFDC1", 1), ("200876_s_at", "PSMB1", -
1),
    ("210019_at", "CALML3", -1), ("221348_at", "NPPC", -1), ("206594_at", "PASK", 1),
    ("219445_at", "GLTSCR1", 1), ("32088_at", "BLZF1", 1), ("216862_s_at", "CMC4", -1),
    ("214277_at", "COX11", 1), ("204726_at", "CDH13", 1), ("214046_at", "FUT9", -1),
    ("202053_s_at", "ALDH3A2", -1), ("218787_x_at", "CWF19L1", -1), ("219756_s_at",
"POF1B", 1),
]
ALIAS = {"GLTSCR1": "BICRA"}

def get(url, data=None):
    req = urllib.request.Request(url, data=data, headers={"Accept": "application/json",
"Content-Type":
"application/json"})
    with urllib.request.urlopen(req, timeout=60) as r:
        return json.load(r)

def main():
    # 1. Entrez IDs
    entrez = {}
    for probe, gene, sign in PANEL:
        sym = ALIAS.get(gene, gene)
        if gene not in entrez:
            hits = [g for g in get(f"{API}/genes?keyword={sym}") if g["hugoGeneSymbol"] ==
sym]
            assert hits, f"gene not found: {sym}"
            entrez[gene] = hits[0]["entrezGeneId"]
            time.sleep(0.2)
    print("entrez:", entrez, flush=True)

    # 2. Sample list for the mrna profile
    sample_ids = get(f"{API}/sample-lists/brca_metabric_mrna/sample-ids")
    print("samples:", len(sample_ids), flush=True)

    # 3. mRNA values for panel genes (chunked sample fetch)
    expr = {g: {} for g in entrez}
    ids = list(entrez.values())
    B = 400
    for i in range(0, len(sample_ids), B):
        chunk = sample_ids[i:i + B]
        body = json.dumps({"entrezGeneIds": ids, "sampleIds": chunk}).encode()

```

```

        rows = get(f"{API}/molecular-profiles/brca_metabric_mrna/molecular-data/fetch",
body)
    for r in rows:
        g = next(k for k, v in entrez.items() if v == r["entrezGeneId"])
        expr[g][r["sampleId"]] = r["value"]
        print(f"fetched {i + len(chunk)}/{len(sample_ids)}", flush=True)
        time.sleep(0.3)

    # 4. Clinical RFS (patient-level, chunked; sample ids == patient ids in this study)
    clin = []
    for i in range(0, len(sample_ids), 400):
        body = json.dumps({"attributeIds": ["RFS_STATUS", "RFS_MONTHS"],
            "ids": sample_ids[i:i + 400]}).encode()
        clin += get(f"{API}/studies/brca_metabric/clinical-data/fetch?
clinicalDataType=PATIENT&projection=SUMMARY", body)
        time.sleep(0.3)
    pat_rfs = {}
    for c in clin:
        pat_rfs.setdefault(c["patientId"], {})[c["clinicalAttributeId"]] = c["value"]
    samp2pat = {s["sampleId"]: s["patientId"]}
        for s in get(f"{API}/studies/brca_metabric/samples?
projection=SUMMARY&pageSize=100000"))
        np.save(OUT / "metabric_sample_ids.npy", np.array(sample_ids))
        json.dump({"entrez": entrez, "expr": expr, "pat_rfs": pat_rfs, "samp2pat": samp2pat},
            open(OUT / "metabric_raw.json", "w"))
        print("pulled:", {g: len(v) for g, v in expr.items()}, "patients with RFS:",
len(pat_rfs), flush=True)

if __name__ == "__main__":
    main()

```

L. src/replication/score_metabric.py

```

"""Locked scoring of the frozen 15-gene panel on METABRIC (PREREG_REPLICATION.md
@e54d8bc).
score_i = sum_g sign_g * z(expr_ig); primary test: directional AUC vs RFS_STATUS,
one-sided 95% bootstrap CI (patient-level, 2000 resamples); sign concordance per gene.
No re-estimation on METABRIC data. Run AFTER pull_metabric.py.
"""
import json
from pathlib import Path

import numpy as np

ROOT = Path(__file__).resolve().parent.parent.parent
OUT = ROOT / "results" / "replication"

PANEL = [ # (probe, gene, discovery sign) - frozen, PREREG section 1
    ("222013_x_at", "EEF2KMT", -1), ("219478_at", "WFDC1", 1), ("200876_s_at", "PSMB1", -
1),
    ("210019_at", "CALML3", -1), ("221348_at", "NPPC", -1), ("206594_at", "PASK", 1),
    ("219445_at", "GLTSCR1", 1), ("32088_at", "BLZF1", 1), ("216862_s_at", "CMC4", -1),
    ("214277_at", "COX11", 1), ("204726_at", "CDH13", 1), ("214046_at", "FUT9", -1),
    ("202053_s_at", "ALDH3A2", -1), ("218787_x_at", "CWF19L1", -1), ("219756_s_at",
"POF1B", 1),
]

```

```

def auc(scores, labels):
    order = np.argsort(scores, kind="mergesort")
    ranks = np.empty(len(scores))
    ranks[order] = np.arange(1, len(scores) + 1)
    # tie-average ranks
    s_sorted = scores[order]
    i = 0
    while i < len(s_sorted):
        j = i
        while j + 1 < len(s_sorted) and s_sorted[j + 1] == s_sorted[i]:
            j += 1
        if j > i:
            ranks[order[i:j + 1]] = (i + 1 + j + 1) / 2
        i = j + 1
    pos = labels == 1
    n_pos, n_neg = pos.sum(), (~pos).sum()
    return (ranks[pos].sum() - n_pos * (n_pos + 1) / 2) / (n_pos * n_neg)

def main():
    raw = json.load(open(OUT / "metabric_raw.json"))
    expr, pat_rfs, samp2pat = raw["expr"], raw["pat_rfs"], raw["samp2pat"]
    sample_ids = list(np.load(OUT / "metabric_sample_ids.npy"))
    samples = [s for s in sample_ids if s in samp2pat]
    labels = np.array([1 if pat_rfs.get(samp2pat[s], {}).get("RFS_STATUS",
    "").startswith("1")
    else 0 if pat_rfs.get(samp2pat[s], {}).get("RFS_STATUS",
    "").startswith("0")
    else -1 for s in samples])
    keep = labels >= 0
    samples = [s for s, k in zip(samples, keep) if k]
    labels = labels[keep]
    X = np.zeros((len(samples), len(PANEL)))
    dropped = []
    for j, (probe, gene, sign) in enumerate(PANEL):
        v = np.array([expr[probe].get(s, np.nan) for s in samples], dtype=float)
        if np.isnan(v).all():
            dropped.append(gene)
            continue
        mu, sd = np.nanmean(v), np.nanstd(v)
        X[:, j] = np.where(np.isnan(v), 0.0, (v - mu) / (sd if sd > 0 else 1.0)) * sign
    score = X.sum(axis=1)
    a = auc(score, labels)
    rng = np.random.default_rng(2026)
    boots = np.array([auc(score[rng.integers(0, len(score), len(score))],
    labels[rng.integers(0, len(labels), len(labels))]) for _ in
    range(2000)])
    # patient-level bootstrap: resample indices iid (patients == samples here, 1
    tumour/patient)
    lo = np.percentile(boots, 5) # one-sided 95%
    conc = []
    for j, (probe, gene, sign) in enumerate(PANEL):
        if gene in dropped:
            continue
        r = np.corrcoef(X[:, j] * sign, labels)[0, 1] # raw-gene direction vs label
        conc.append((gene, sign, float(r), bool(np.sign(r) == sign)))
    n_match = sum(c[3] for c in conc)
    res = {
        "cohort": "METABRIC brca_metabric (cBioPortal public API, pulled 2026-09-27)",

```

```

        "n_samples": len(samples), "n_relapsed": int(labels.sum()),
        "genes_used": len(PANEL) - len(dropped), "genes_dropped_from_profile": dropped,
        "primary_auc": float(a), "boot_lo_one_sided_95": float(lo),
        "sign_concordance": f"{n_match}/{len(conc)}",
        "per_gene": [{"gene": g, "discovery_sign": s, "metabric_corr": r, "concordant": m}
                      for g, s, r, m in conc],
        "primary_test_pass": bool(lo > 0.5 and n_match / len(conc) > 0.5),
        "gate": "PREREG_REPLICATION.md section 3-4",
    }
    json.dump(res, open(OUT / "metabric_primary.json", "w"), indent=1)
    print(json.dumps({k: v for k, v in res.items() if k != "per_gene"}, indent=1))
    for c in conc:
        print(c)

```

```

if __name__ == "__main__":
    main()

```

L. src/replication/score_scanb.py

```

"""SCAN-B (GSE96058) locked scoring - SECONDARY arm, PREREG_REPLICATION.md section 2/3.
GEO clinical annotation carries OS only -> declared OS-endpoint arm (prereg fallback),
NOT pooled with RFS arms. Same locked rule: score = sum sign_g * z(expr), no re-
estimation.
"""

```

```

import csv, gzip, json
from pathlib import Path

```

```

import numpy as np

```

```

ROOT = Path(__file__).resolve().parent.parent.parent
DAT = ROOT / "data" / "scanb"
OUT = ROOT / "results" / "replication"
OUT.mkdir(parents=True, exist_ok=True)

```

```

# (matrix row name, discovery sign); EEF2KMT=FAM86A, BICRA=GLTSCR1 (REPLICATION_ACCESS.md)
PANEL = [("EEF2KMT", "FAM86A", -1), ("WFDC1", "WFDC1", 1), ("PSMB1", "PSMB1", -1),
          ("CALML3", "CALML3", -1), ("NPPC", "NPPC", -1), ("PASK", "PASK", 1),
          ("GLTSCR1", "GLTSCR1", 1), ("BLZF1", "BLZF1", 1), ("CMC4", "CMC4", -1),
          ("COX11", "COX11", 1), ("CDH13", "CDH13", 1), ("FUT9", "FUT9", -1),
          ("ALDH3A2", "ALDH3A2", -1), ("CWF19L1", "CWF19L1", -1), ("POF1B", "POF1B", 1)]

```

```

def parse_matrix(path):
    clin = {}
    with gzip.open(path, "rt") as f:
        titles = None
        for line in f:
            if line.startswith("!Sample_title"):
                titles = [t.strip('"') for t in line.rstrip("\n").split("\t")[1:]]
                for t in titles:
                    clin[t] = {}
            elif line.startswith("!Sample_characteristics_ch1") and titles:
                vals = line.rstrip("\n").split("\t")[1:]
                for t, v in zip(titles, vals):
                    v = v.strip('"')
                    if ":" in v:
                        k, val = v.split(": ", 1)
                        clin[t][k.lower()] = val

```

```
return clin
```

```
def auc(scores, labels):
    order = np.argsort(scores, kind="mergesort")
    ranks = np.empty(len(scores))
    ranks[order] = np.arange(1, len(scores) + 1)
    s_sorted = scores[order]
    i = 0
    while i < len(s_sorted):
        j = i
        while j + 1 < len(s_sorted) and s_sorted[j + 1] == s_sorted[i]:
            j += 1
        if j > i:
            ranks[order[i:j + 1]] = (i + 1 + j + 1) / 2
        i = j + 1
    pos = labels == 1
    return (ranks[pos].sum() - pos.sum() * (pos.sum() + 1) / 2) / (pos.sum() *
(~pos).sum())
```

```
def main():
    clin = {}
    for f in ["GSE96058-GPL11154_series_matrix.txt.gz", "GSE96058-
GPL18573_series_matrix.txt.gz"]:
        clin.update(parse_matrix(DAT / f))
    with open(DAT / "panel_rows.csv") as f:
        rows = list(csv.reader(f))
    cols = rows[0][1:]
    expr = {r[0].strip(''): np.array([float(x) for x in r[1:]] for r in rows[1:])}
    keep_cols, labels = [], []
    for i, c in enumerate(cols):
        rec = clin.get(c)
        if rec and rec.get("overall survival event", "NA") not in ("NA", ""):
            keep_cols.append(i)
            labels.append(int(float(rec["overall survival event"])))
    labels = np.array(labels)
    X = np.zeros((len(keep_cols), len(PANEL)))
    for j, (gene, rowname, sign) in enumerate(PANEL):
        v = expr[rowname][keep_cols]
        mu, sd = v.mean(), v.std()
        X[:, j] = (v - mu) / (sd if sd > 0 else 1.0) * sign
    score = X.sum(axis=1)
    a = auc(score, labels)
    rng = np.random.default_rng(2026)
    n = len(score)
    boots = []
    for _ in range(2000):
        idx = rng.integers(0, n, n)
        if labels[idx].sum() in (0, n):
            continue
        boots.append(auc(score[idx], labels[idx]))
    lo = float(np.percentile(boots, 5))
    conc = []
    for j, (gene, rowname, sign) in enumerate(PANEL):
        r = float(np.corrcoef(X[:, j] * sign, labels)[0, 1])
        conc.append({"gene": gene, "discovery_sign": sign, "scanb_corr": r,
"concordant": bool(np.sign(r) == sign)})
    n_match = sum(c["concordant"] for c in conc)
    res = {
```

```

        "cohort": "SCAN-B GSE96058 (GEO public, pulled 2026-09-27; RNA-seq, both
platforms)",
        "endpoint": "OS (overall survival event) - DECLARED OS-endpoint arm; GEO
annotation carries no relapse field (prereg section 2 fallback)",
        "n_samples": int(n), "n_events": int(labels.sum()),
        "genes_used": len(PANEL),
        "auc": float(a), "boot_lo_one_sided_95": lo,
        "sign_concordance": f"{n_match}/{len(PANEL)}",
        "per_gene": conc,
        "secondary_test_pass": bool(lo > 0.5 and n_match / len(PANEL) > 0.5),
        "gate": "PREREG_REPLICATION.md section 3 (SECONDARY 1)",
    }
    json.dump(res, open(OUT / "scanb_secondary.json", "w"), indent=1)
    print(json.dumps({k: v for k, v in res.items() if k != "per_gene"}, indent=1))

```

```

if __name__ == "__main__":
    main()

```

L. src/replication/twocohort_selection.py

```

"""Two-cohort cross-stability selection (PREREG_TWOCOHORT.md section 2, locked @b476288
BEFORE this ran). Discovery cohorts ONLY: GSE2034 (relapse) + METABRIC (RFS).
Recipe per cohort (identical to experiments/gene_stability.py): z-score, top-2000
symbols by |t|, 40 bootstrap L1 logistic fits (C=0.5, max_iter=500,
StratifiedKFold(5, shuffle, seed 0) rotating splits, bootstrap resample, rng seed 0),
stability = fraction of fits keeping majority sign. Panel = stability>=0.975 in BOTH
cohorts AND majority signs agree. No test-cohort data is touched here.
Usage: twocohort_selection.py [gse2034|metabric|panel]
"""

```

```

import json, sys
from pathlib import Path

```

```

import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold

```

```

ROOT = Path(__file__).resolve().parent.parent.parent
OUT = ROOT / "results" / "replication"
OUT.mkdir(parents=True, exist_ok=True)

```

```

def tstats_chunked(X, y, step=2000):
    t = np.zeros(X.shape[1], dtype=np.float32)
    for i in range(0, X.shape[1], step):
        c = np.nan_to_num(X[:, i:i + step].astype(np.float32))
        sd = c.std(0) + 1e-8
        t[i:i + step] = (c[y == 1].mean(0) - c[y == 0].mean(0)) / sd
    del c
    return t

```

```

def stability_run(X, y, names):
    t = tstats_chunked(X, y)
    idx = np.argsort(-np.abs(t))[:2000]
    idx.sort()
    Xs = np.nan_to_num(X[:, idx].astype(np.float32))
    Xs = (Xs - Xs.mean(0)) / (Xs.std(0) + 1e-8)

```

```

gsel = [names[i] for i in idx]
rng = np.random.RandomState(0)
skf = StratifiedKFold(5, shuffle=True, random_state=0)
folds = list(skf.split(Xs, y))
coefs = []
for b in range(40):
    tr, _ = folds[b % 5]
    sub = rng.choice(tr, size=len(tr), replace=True)
    clf = LogisticRegression(max_iter=500, C=0.5).fit(Xs[sub], y[sub])
    coefs.append(clf.coef_[0])
    if (b + 1) % 10 == 0:
        print(f" {b+1}/40", flush=True)
C = np.array(coefs)
sign = np.sign(C.mean(0))
stability = (np.sign(C) == sign).mean(0)
return {g: {"stability": float(s), "sign": int(sg), "mean_coef": float(c)}
        for g, s, sg, c in zip(gsel, stability, sign, C.mean(0))}

def gse2034():
    sys.path.insert(0, str(ROOT / "src"))
    from biomedml.io_geo import parse_series_matrix
    X, probes, meta = parse_series_matrix(str(ROOT / "data" / "GSE2034_matrix.txt.gz"))
    key = [k for k in meta if "relapse" in k.lower() or "metasta" in k.lower() or "event"
in k.lower()]
    y = np.array([1 if v.strip().lower() in ("1", "yes", "true") else 0 for v in
meta[key[0]]])
    p2s = json.load(open(ROOT / "results" / "u133a_probe_to_symbol.json"))
    X = X.astype(np.float32)
    t = np.abs(tstats_chunked(X, y))
    # max-|t| probe per symbol
    best = {}
    for i, p in enumerate(probes):
        s = p2s.get(p)
        if s and (s not in best or t[i] > best[s][1]):
            best[s] = (i, t[i])
    idxs = sorted(v[0] for v in best.values())
    names = [p2s[probes[i]] for i in idxs]
    print("symbols:", len(names), "samples:", X.shape[0], flush=True)
    res = stability_run(X[:, idxs], y, names)
    json.dump(res, open(OUT / "sel_gse2034.json", "w"))
    print("gse2034 arm done:", len(res), flush=True)

def metabric():
    z = np.load(OUT / "metabric_matrix.npz", allow_pickle=True)
    mat, gene_ids, sample_ids = z["mat"], list(z["gene_ids"]), list(z["sample_ids"])
    raw = json.load(open(OUT / "metabric_raw.json"))
    pat_rfs, samp2pat = raw["pat_rfs"], raw["samp2pat"]
    idxj = json.load(open(OUT / "mb_index_symbols.json")) if (OUT /
"mb_index_symbols.json").exists() else None
    # symbols from index.json were saved in chunk dir (deleted); rebuild from pull index
copy
    sym_map = json.load(open(OUT / "mb_symbols.json"))
    labels = np.array([1 if pat_rfs.get(samp2pat.get(s, ""), {}).get("RFS_STATUS",
"" ).startswith("1")
else 0 if pat_rfs.get(samp2pat.get(s, ""), {}).get("RFS_STATUS",
"" ).startswith("0")
else -1 for s in sample_ids])
    keep = labels >= 0

```

```

X = mat[:, keep].T
y = labels[keep]
names = [sym_map.get(str(g), str(g)) for g in gene_ids]
# dedupe symbols: keep max-|t| (chunked, no full-size copies)
t = np.abs(tstats_chunked(X, y))
best = {}
for i, s in enumerate(names):
    if s and (s not in best or t[i] > best[s][1]):
        best[s] = (i, t[i])
idxs = sorted(v[0] for v in best.values())
names2 = [names[i] for i in idxs]
print("symbols:", len(names2), "samples:", X.shape[0], "events:", int(y.sum()),
flush=True)
res = stability_run(X[:, idxs], y, names2)
json.dump(res, open(OUT / "sel_metabric.json", "w"))
print("metabric arm done:", len(res), flush=True)

def panel():
    a = json.load(open(OUT / "sel_gse2034.json"))
    b = json.load(open(OUT / "sel_metabric.json"))
    genes = []
    for g in set(a) & set(b):
        if a[g]["stability"] >= 0.975 and b[g]["stability"] >= 0.975 and a[g]["sign"] ==
b[g]["sign"]:
            genes.append({"gene": g, "sign": a[g]["sign"],
                          "stab_gse2034": round(a[g]["stability"], 3),
                          "stab_metabric": round(b[g]["stability"], 3)})
    genes.sort(key=lambda r: -min(r["stab_gse2034"], r["stab_metabric"]))
    if len(genes) > 30:
        genes = genes[:30]
    json.dump({"rule": "stability>=0.975 both cohorts + sign agreement
(PREREG_TWOCOHORT.md)",
              "n_panel": len(genes), "panel": genes},
              open(OUT / "twocohort_panel.json", "w"), indent=1)
    print("PANEL:", len(genes))
    for g in genes:
        print(g)

if __name__ == "__main__":
    {"gse2034": gse2034, "metabric": metabric, "panel": panel}[sys.argv[1]]()

```

Appendix D. Source listings

D. src/biomedml/io_geo.py

```

"""Parse a GEO series-matrix file (GSEnnn_series_matrix.txt.gz)."""
from __future__ import annotations
import gzip
import numpy as np

def parse_series_matrix(path):
    """Returns (X [samples x genes], gene_ids, meta dict of column->values)."""
    opener = gzip.open if path.endswith(".gz") else open
    meta = {}
    header = None

```

```

rows = []
with opener(path, "rt", errors="replace") as f:
    for line in f:
        if line.startswith("!Sample_characteristics"):
            parts = line.rstrip("\n").split("\t")
            cells = [c.strip(' ') for c in parts[1:]]
            # Key from the first non-empty cell's "key:" prefix: GEO files
            # like GSE2603 leave the first samples' cells blank on some
            # characteristic lines, so parts[1] can be empty.
            key = ""
            for c in cells:
                if ":" in c:
                    key = c.split(":", 1)[0].strip()
                    break
            if not key:
                continue
            vals = [c.split(":", 1)[-1].strip() if ":" in c else "" for c in cells]
            meta.setdefault(key, vals)
        elif line.startswith("!Sample_geo_accession"):
            header = [p.strip(' ') for p in line.rstrip("\n").split("\t")[1:]]
        elif line.startswith("!series_matrix_table_begin"):
            cols = next(f).rstrip("\n").split("\t")
            genes = []
            data = []
            for l2 in f:
                if l2.startswith("!series_matrix_table_end"):
                    break
                p = l2.rstrip("\n").split("\t")
                genes.append(p[0].strip(' '))
                data.append([float(x) if x not in ("", "null", "NA") else float("nan")
                             for x in p[1:]])
            X = np.array(data, dtype=np.float32).T # samples x genes
            return X, genes, meta
    raise ValueError("no series matrix table found")

```

D. src/biomedml/stats.py

```

"""Differential expression and multiple-testing control."""
from __future__ import annotations
import numpy as np
from scipy import stats as sps

def de_genes(X, y, max_genes=None):
    """Welch t-test per gene; returns arrays (t, p) aligned to genes."""
    X0, X1 = X[y == 0], X[y == 1]
    t, p = sps.ttest_ind(X1, X0, axis=0, equal_var=False, nan_policy="omit")
    return np.nan_to_num(t), np.nan_to_num(p, nan=1.0)

def bh_fdr(p, alpha=0.05):
    """Benjamini-Hochberg; returns boolean mask of discoveries."""
    p = np.asarray(p)
    n = len(p)
    order = np.argsort(p)
    ranked = p[order]
    thresh = alpha * np.arange(1, n + 1) / n
    below = ranked <= thresh
    if not below.any():
        return np.zeros(n, bool)

```

```

kmax = np.max(np.where(below))
mask = np.zeros(n, bool)
mask[order[:kmax + 1]] = True
return mask

```

```

def coexpression_graph(X, top_idx, thresh=0.6):
    """Gene co-expression adjacency over selected genes (|r| >= thresh)."""
    Z = X[:, top_idx]
    Z = (Z - Z.mean(0)) / (Z.std(0) + 1e-8)
    R = (Z.T @ Z) / Z.shape[0]
    A = (np.abs(R) >= thresh).astype(np.float32)
    np.fill_diagonal(A, 0.0)
    return A, R

```

D. src/biomedml/models.py

```

"""1D-CNN over expression profiles + logistic baseline (numpy)."""
from __future__ import annotations
import numpy as np
import torch
import torch.nn as nn

class ExprCNN(nn.Module):
    def __init__(self, n_genes, n_classes=2, hidden=32):
        super().__init__()
        # treat the (sorted) expression vector as a 1-channel signal
        self.conv = nn.Sequential(
            nn.Conv1d(1, hidden, 25, stride=5, padding=12), nn.ReLU(),
            nn.Conv1d(hidden, hidden, 15, stride=5, padding=7), nn.ReLU(),
            nn.AdaptiveAvgPool1d(8), nn.Flatten())
        self.head = nn.Linear(hidden * 8, n_classes)

    def forward(self, X): # (B, G)
        return self.head(self.conv(X.unsqueeze(1)))

def train_torch(model, X, y, tr, epochs=20, lr=1e-3, batch=32, seed=0):
    torch.manual_seed(seed)
    opt = torch.optim.Adam(model.parameters(), lr=lr)
    lossf = nn.CrossEntropyLoss()
    Xt = torch.tensor(X, dtype=torch.float32)
    yt = torch.tensor(y, dtype=torch.long)
    for _ in range(epochs):
        perm = torch.randperm(len(tr))
        for i in range(0, len(tr), batch):
            b = torch.tensor(tr)[perm[i:i + batch]]
            opt.zero_grad()
            loss = lossf(model(Xt[b]), yt[b])
            loss.backward()
            opt.step()
    return model

def predict_torch(model, X):
    model.eval()
    with torch.no_grad():

```

```

Xt = torch.tensor(X, dtype=torch.float32)
return model(Xt).argmax(1).numpy()

class LogRegNP:
    def __init__(self, n_in, lr=0.1, epochs=500, l2=1e-3, seed=0):
        self.W = np.random.default_rng(seed).normal(0, 0.01, (n_in, 2))
        self.b = np.zeros(2)
        self.lr, self.epochs, self.l2 = lr, epochs, l2

    def fit(self, X, y):
        Y = np.eye(2)[y]
        for _ in range(self.epochs):
            Z = X @ self.W + self.b
            Z -= Z.max(1, keepdims=True)
            P = np.exp(Z) / np.exp(Z).sum(1, keepdims=True)
            G = (P - Y) / len(X)
            self.W -= self.lr * (X.T @ G + self.l2 * self.W)
            self.b -= self.lr * G.sum(0)
        return self

    def predict(self, X):
        return (X @ self.W + self.b).argmax(1)

def auc_score(y_true, scores):
    """Rank-based AUC (Mann-Whitney)."""
    y_true = np.asarray(y_true); scores = np.asarray(scores)
    pos = scores[y_true == 1]; neg = scores[y_true == 0]
    if len(pos) == 0 or len(neg) == 0:
        return float("nan")
    wins = sum((p > neg).sum() + 0.5 * (p == neg).sum() for p in pos)
    return float(wins / (len(pos) * len(neg)))

```

D. experiments/gene_stability.py

```

"""Gene-stability arm: which genes robustly carry the metastasis signal?

Bootstrap the 5-fold-CV logistic pipeline 40 times; a gene's stability is
the fraction of fits where its coefficient keeps the majority sign, times
mean |coef|. Outputs a ranked, falsifiable stable-panel candidate list.
"""

import json, os, sys
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold
sys.path.insert(0, "src")
from biomedml.io_geo import parse_series_matrix

def main():
    X, genes, meta = parse_series_matrix("data/GSE2034_matrix.txt.gz")
    key = [k for k in meta if "relapse" in k.lower() or "metasta" in k.lower() or "event"
in k.lower()]
    y = np.array([1 if v.strip().lower() in ("1", "yes", "true") else 0 for v in
meta[key[0]]])
    X = np.nan_to_num(X)
    X = (X - X.mean(0)) / (X.std(0) + 1e-8)
    # prefilter to top 2000 by |t| to bound compute (same as pipeline)

```

```

from biomedml.stats import de_genes
t, _ = de_genes(X, y)
idx = np.argsort(-np.abs(t))[:2000]
Xs, gsel = X[:, idx], [genes[i] for i in idx]

rng = np.random.RandomState(0)
B, coefs = 40, []
skf = StratifiedKFold(5, shuffle=True, random_state=0)
folds = list(skf.split(Xs, y))
for b in range(B):
    tr, _ = folds[b % 5]
    sub = rng.choice(tr, size=len(tr), replace=True)
    clf = LogisticRegression(max_iter=500, C=0.5).fit(Xs[sub], y[sub])
    coefs.append(clf.coef_[0])
C = np.array(coefs) # B x G
sign = np.sign(C.mean(0))
stability = (np.sign(C) == sign).mean(0)
score = stability * np.abs(C.mean(0))
order = np.argsort(-score)[:30]
panel = [dict(probe=gsel[i], stability=round(float(stability[i]), 3),
              mean_coef=round(float(C.mean(0)[i]), 4),
              direction="risk" if sign[i] > 0 else "protective") for i in order]
os.makedirs("results", exist_ok=True)
json.dump(dict(n_boot=40, n_genes_prefilter=2000, panel=panel),
          open("results/gene_stability.json", "w"), indent=1)
for p in panel[:15]:
    print(p)

if __name__ == "__main__":
    main()

```